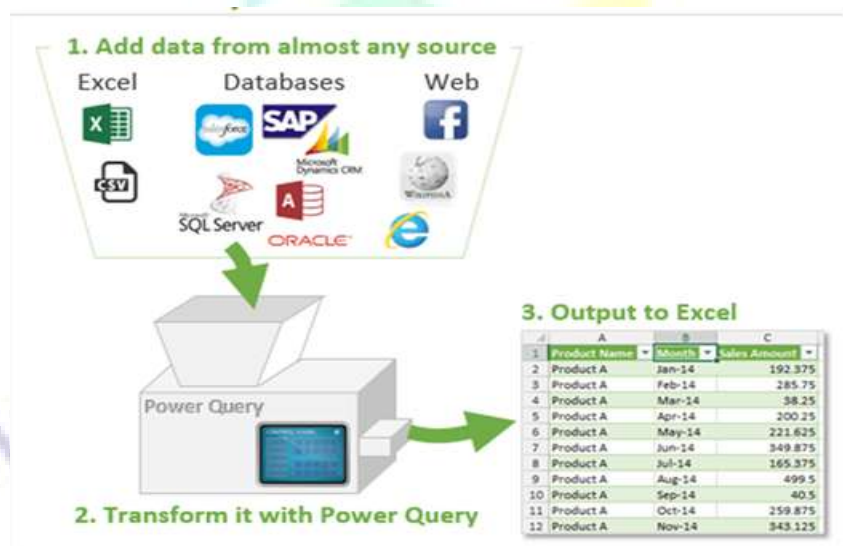


Module-02**RBT Level-L1, L2**

Power Query –Data types and Transformations, Power Query working with data types, Filter the data, PBI Filters, Column and Row Transformation, Column and Row Transforming, Row Transformations, Combine Queries.

Introduction to Power Query

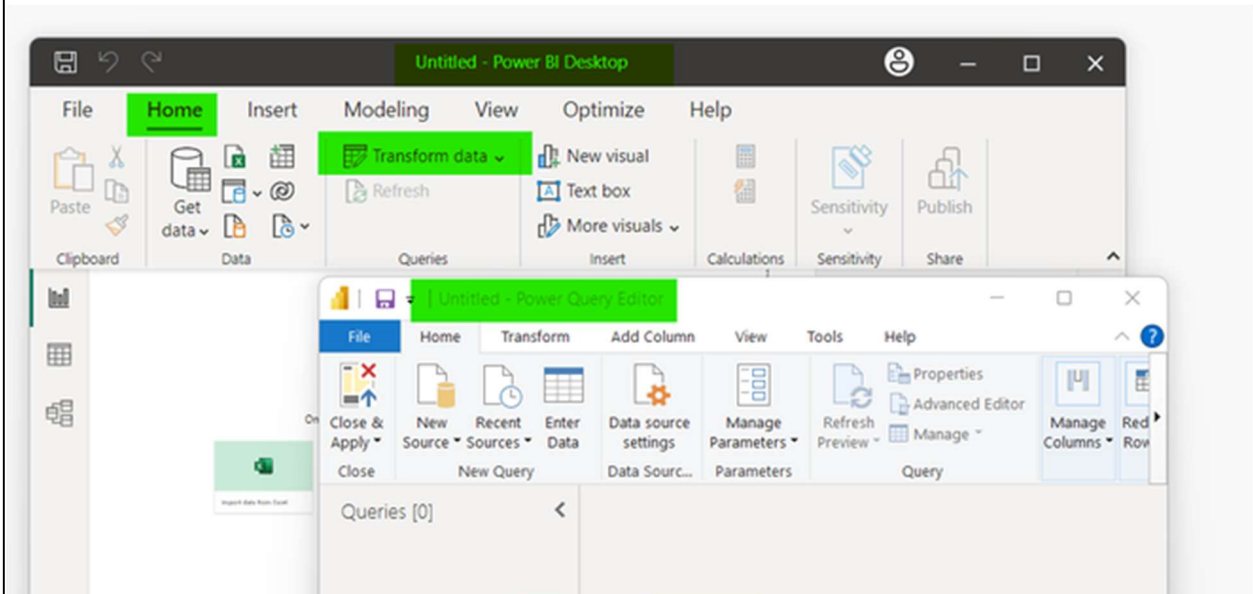
Power Query in Power BI is a powerful data transformation and cleansing engine (ETL—Extract, Transform, Load) that allows you to connect to, reshape, and clean data from hundreds of sources. Accessible via "Transform Data," it enables users to automate, merge, and format data without coding, using a formula language called M.

Working of Power Query

- ❖ Add data sources (Excel tables, CSV files, database tables, webpages, etc.)
- ❖ Press buttons in the Power Query Editor window to transform the data.
- ❖ Output that data to worksheet or data model (PowerPivot) that is ready for pivot tables or reporting.

Launching Power Query Editor

To open the Power Query Editor click **Transform Data** under the **Home** tab in Power BI Desktop. This will open the editor where you can load and transform data.



Now we will add data to query editor to perform desired operations.

Importing Data

In the Power Query Editor click **New Source** to add data from different sources like Excel, SQL, or the web.

After importing we can perform necessary transformations on the data.

The data referred to here for illustration purposes is an Excel file named **My movie list.xlsx** and its data is as follows:

My Movie List ☆ 📁 ☁

File Edit View Insert Format Data Tools Extensions Help

↶ ↷ 🖨 🗑 100% | \$ % .0 .00 123 | Arial |

D17 | fx 8.6

	A	B	C	D
1	RANK	TITLE	GENRE	RATING
2	1	The Shawshank Redemption	Drama	9.3
3	2	The Godfather	Crime, Drama	9.2
4	3	Schindler's List	Biography, Drama, History	
5	4	Se7en	Crime, Drama, Mystery	8.9
6	5	It's a Wonderful Life	Comedy, Drama, Family	8.6
7	6	Goodfellas	Biography, Crime, Drama	8.7
8	7	Romeo + Juliet	Drama, Romance	6.7
9	8	12 Angry Men		9
10	9	Alien	Horror, Sci-Fi	8.4
11	10	Toy Story	Animation, Adventure, Comedy	8.3
12	11		Drama	8.7
13	12	To Kill a Mockingbird	Crime, Drama	
14	13	Rumble in the Bronx	Action, Comedy, Crime	6.8
15	14		Crime, Drama	7.5
16	15	Psycho	Horror, Mystery, Thriller	8.5
17	16	The Silence of the Lambs	Crime, Drama, Horror	8.6

On loading this data on the query editor it appears as:

	123 RANK	A8C TITLE	A8C GENRE	12 RATING
1		1 The Shawshank Redemption	Drama	9.3
2		2 The Godfather	Crime, Drama	9.2
3		3 Schindler's List	Biography, Drama, History	null
4		4 Se7en	Crime, Drama, Mystery	8.9
5		5 It's a Wonderful Life	Comedy, Drama, Family	8.6
6		6 Goodfellas	Biography, Crime, Drama	8.7
7		7 Romeo + Juliet	Drama, Romance	6.7
8		8 12 Angry Men		9
9		9 Alien	Horror, Sci-Fi	8.4
10		10 Toy Story	Animation, Adventure, Comedy	8.3
11		11 null	Drama	8.7
12		12 To Kill a Mockingbird	Crime, Drama	null
13		13 Rumble in the Bronx	Action, Comedy, Crime	6.8
14		14 null	Crime, Drama	7.5
15		15 Psycho	Horror, Mystery, Thriller	8.5
16		16 The Silence of the Lambs	Crime, Drama, Horror	8.6
17		null	null	null

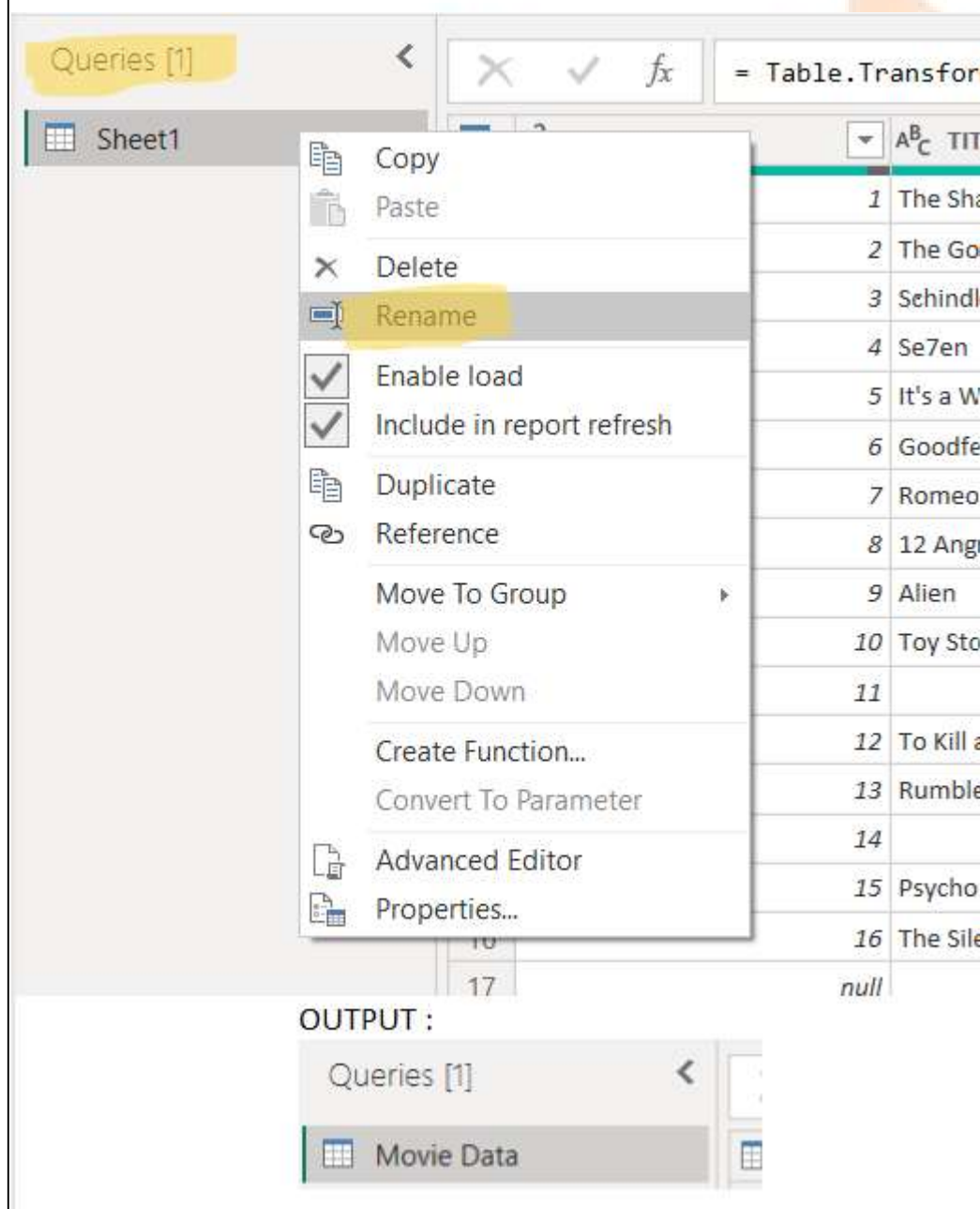
Now since we have the Excel sheet and data imported on the power query editor as well we can perform transformations.

1. Renaming Elements

The power query editor provides us with a variety of possible renaming. We can rename the data sources or tables, columns and queries.

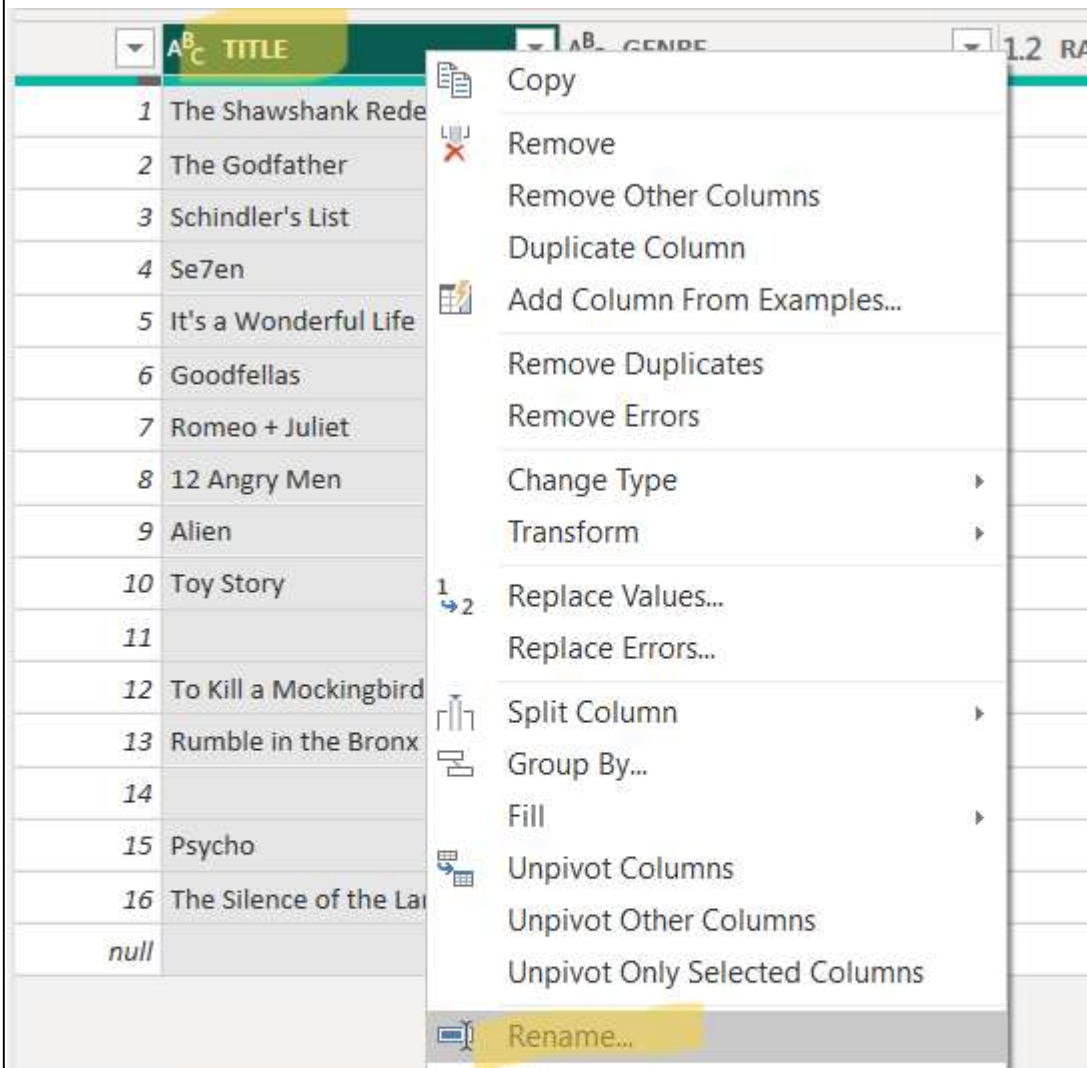
Rename Data Source

It can be achieved by right-clicking on the source and opting for the rename option. Here, we have renamed the data source from sheet 1 to Movie Data. On transformation, we get :



Rename Column

It's also done by right-clicking the column in which we wish to change the name and selecting the option of renaming and renaming as per the wish of the user. Here, we have renamed the column "TITLE" to "MOVIE NAME". On renaming the column it appears as :



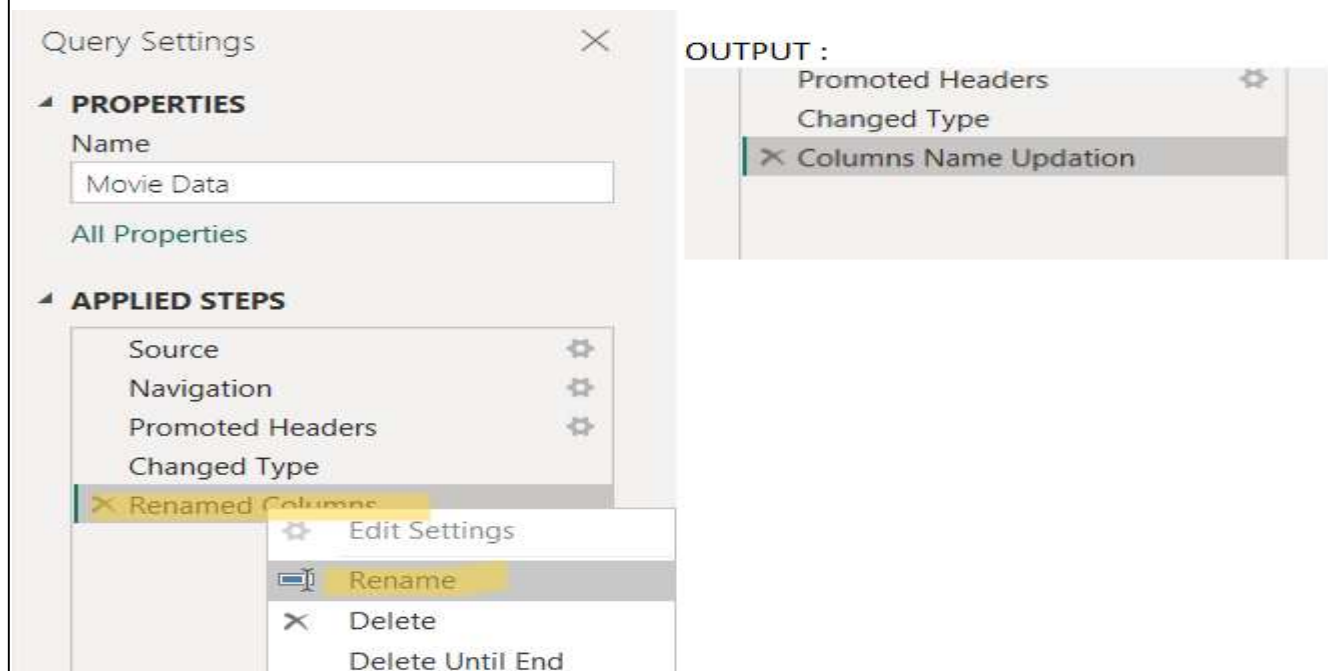
The screenshot shows a data table with columns 'TITLE' and 'GENRE'. A right-click context menu is open over the 'TITLE' column, with the 'Rename...' option highlighted. Below the menu, the 'OUTPUT' shows the resulting table where the 'TITLE' column has been renamed to 'MOVIE NAME'.

	AB C TITLE	AB C GENRE
1	The Shawshank Rede	
2	The Godfather	
3	Schindler's List	
4	Se7en	
5	It's a Wonderful Life	
6	Goodfellas	
7	Romeo + Juliet	
8	12 Angry Men	
9	Alien	
10	Toy Story	
11		
12	To Kill a Mockingbird	
13	Rumble in the Bronx	
14		
15	Psycho	
16	The Silence of the La	
	null	

	AB C MOVIE NAME	AB C GI
1	The Shawshank Redemption	Drama
2	The Godfather	Crime,
3	Schindler's List	Biogra

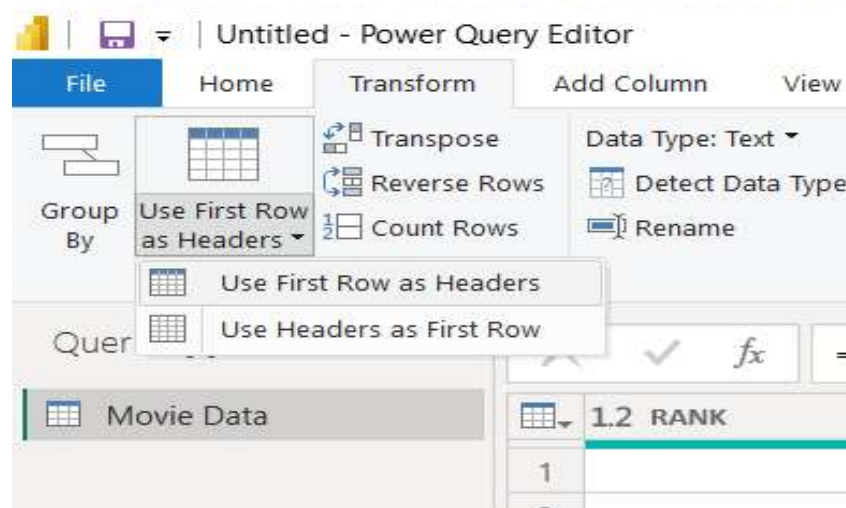
Rename Query

Under the query settings pane an **"Applied Steps"** section under which all the changes we made are stored as queries. Using the query editor we can rename the queries also. Here, we have renamed the query **"Renamed Columns"** to **"Columns Name Updation"**. On renaming it appears as :



2. Make First Row as the Header in Power BI

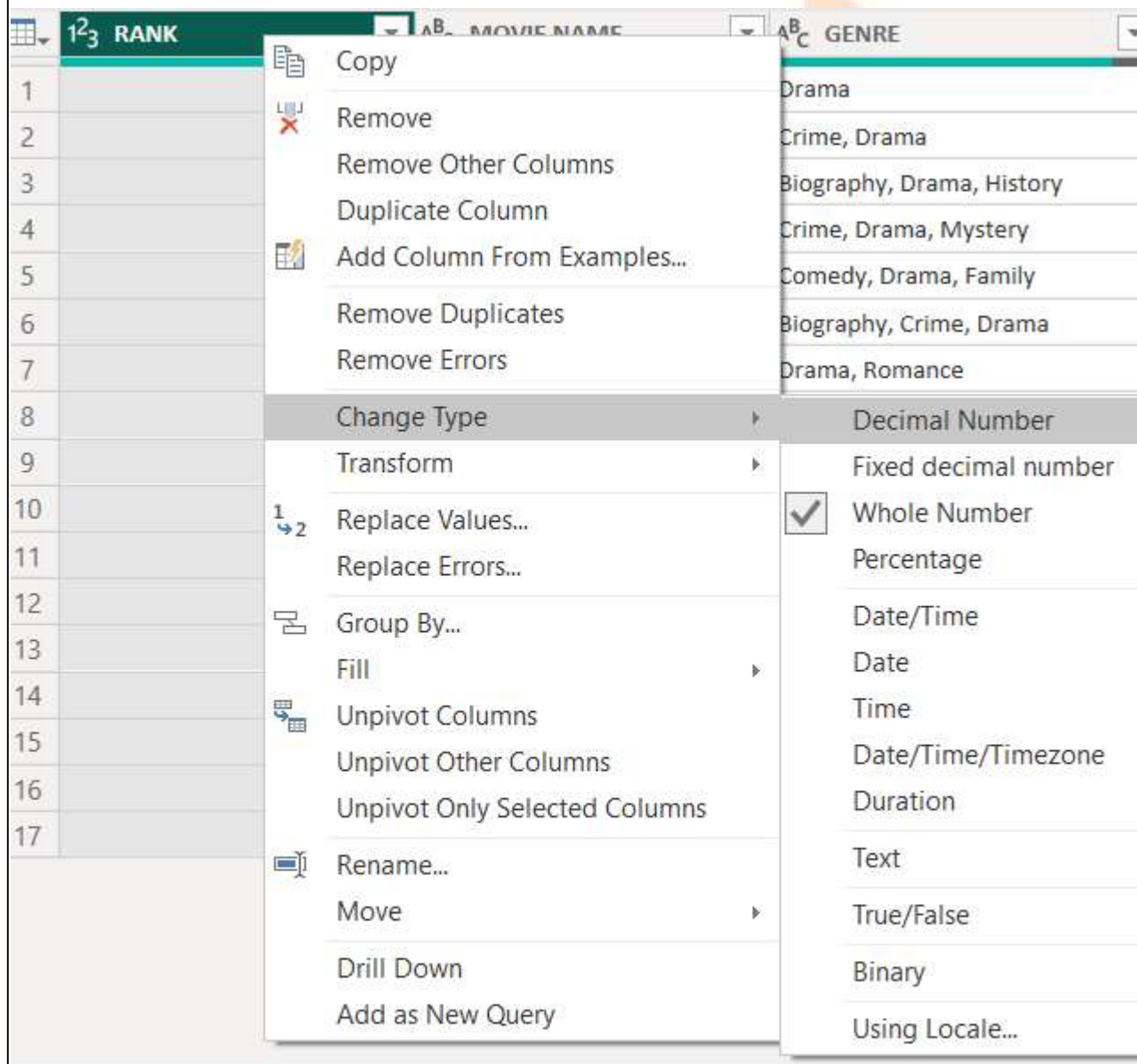
This operation is used to set the first row as the column header. This option is available under the **"TRANSFORM"** of the ribbon. It has another option of **"Use Headers as First Row"** as well.



3. Change Data Types of a Column in Power BI

To change a column's data type in Power BI, right-click the column, select **Change Type** and choose the appropriate data type (e.g., "Whole Number" to "Decimal Number").

For example, we changed the "RANK" column from whole numbers (1, 2, 3) to decimal numbers (1.2, 2.3, 3.4). You can also change data types from the **Home** tab under "Data type"



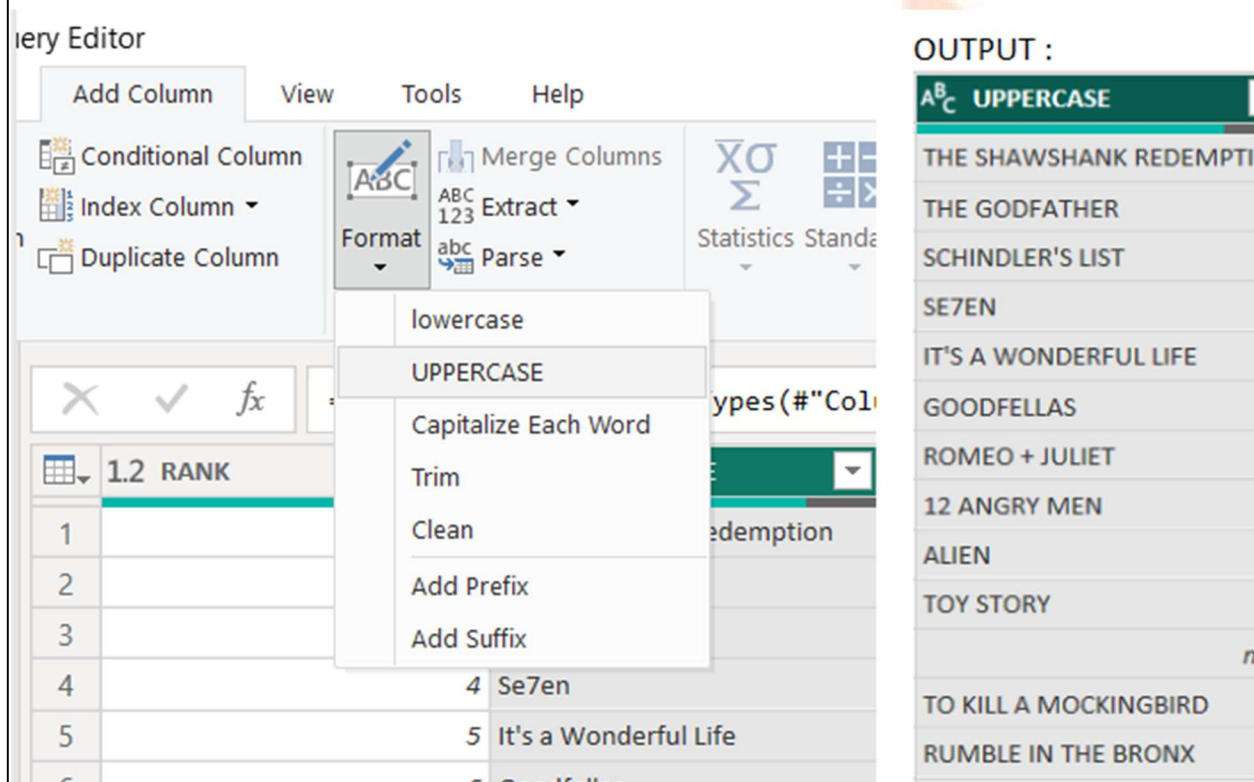
OUTPUT :



4. Format Operation in Power BI

The Format feature in the Transform tab helps format text in Power BI. We can change text to uppercase or lowercase, add prefixes or suffixes and remove leading or trailing spaces using the Trim option.

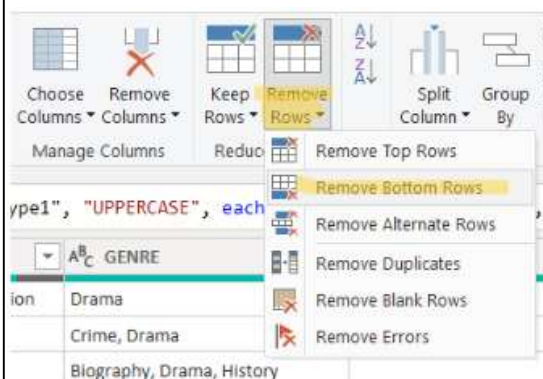
The Clean option removes non-printable characters. To use these simply click the relevant option. In this example we applied the UPPERCASE transformation but the other options work the same way.



5. Removal Operation in Power BI

Remove Row Operation

The **Reduce Rows** feature in the **Home** tab allows to remove rows in Power BI. We can remove top, bottom, or alternate rows, as well as duplicates, blank rows and errors.



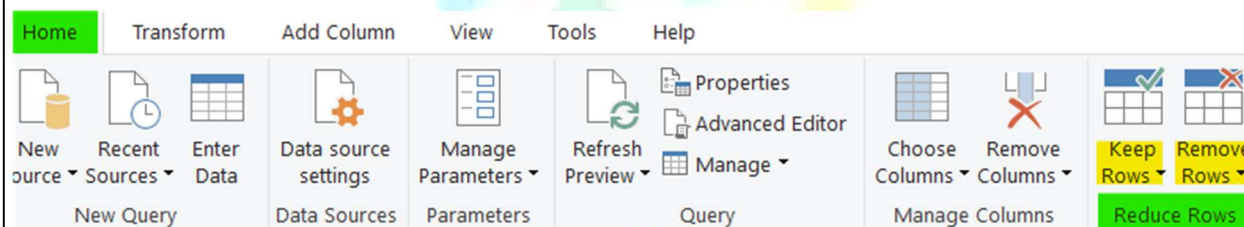
Number of rows to be removed is 1

Row to be removed is number 17 :

16	16	The Silence of the Lambs	Crime, Drama, Horror	8.6	THE SILENCE OF THE LAMBS
17	null	null	null	null	null

Output : Row number 17 has been removed as follows

16	16	The Silence of the Lambs	Crime, Drama, Horror	8.6	THE SILENCE OF THE LAMBS
----	----	--------------------------	----------------------	-----	--------------------------

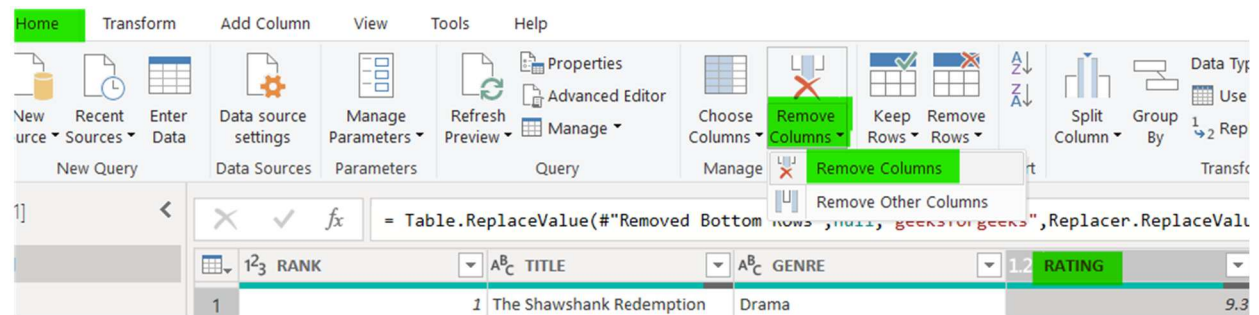


Please note that the "Reduce Rows" block in Power BI offers two options: "Remove Rows" and "Keep Rows".

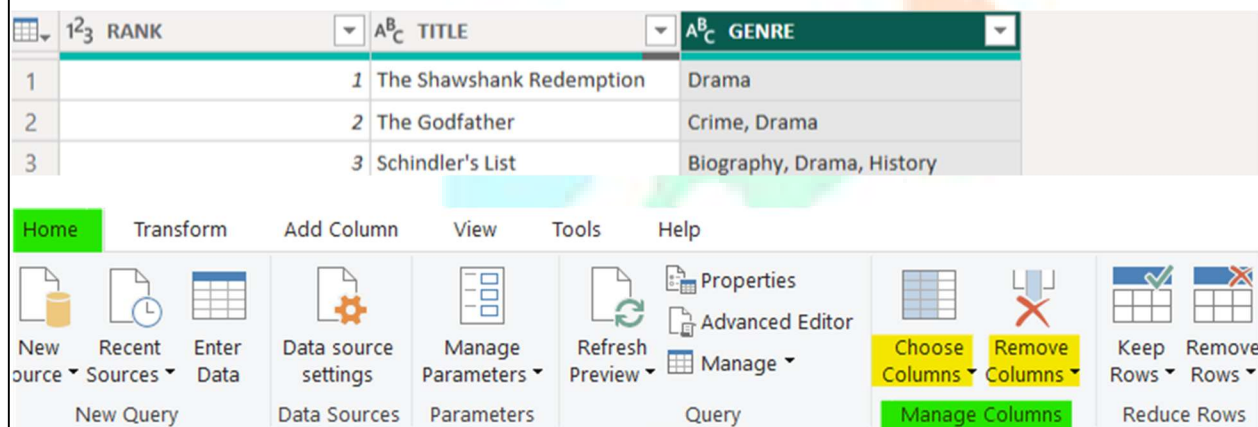
"Remove Rows" operation above. On the same lines the "Keep Rows" operation is performed. The "Keep Rows" feature lets we select the rows we want to keep while "Remove Rows" deletes unwanted rows.

Remove Column Operation

The **Remove Columns** feature allows to delete one or more columns. It's found under the **"Home"** tab in the **Manage Columns** section. we can click "Remove Columns" to delete the selected column or use "Remove Other Columns" to delete all columns except the one you selected. For example if you remove the "RATING" column it will no longer appear in your data.



Output:



The **Manage Columns** block has two features: **Choose Columns** and **Remove Columns**. **Choose Columns** lets you keep the columns you want while **Remove Columns** lets us delete the ones we don't need.

6. Merge Operation in Power BI

We can merge multiple columns. To select multiple columns hold down the Ctrl key, navigate to the columns you want to be selected using the left and right arrows and then press the Space bar to actually select those columns.

The "Add Column" bar supports the "Merge Columns" feature which is followed by a prompt of merge columns that asks for the name of the merged column and to set the separator.

Here, we have merged the columns "GENRE" and "RATING". We have used a custom separator " ;) " and kept the name of the merged column simple as "Merged". The illustration along with the output is as follows :

Software interface showing the 'Merge Columns' dialog box. The dialog box is titled 'Merge Columns' and contains the following options:

- Separator:** --Custom--
- ;) :**
- New column name (optional):** Merged

Buttons: OK, Cancel

Background data table:

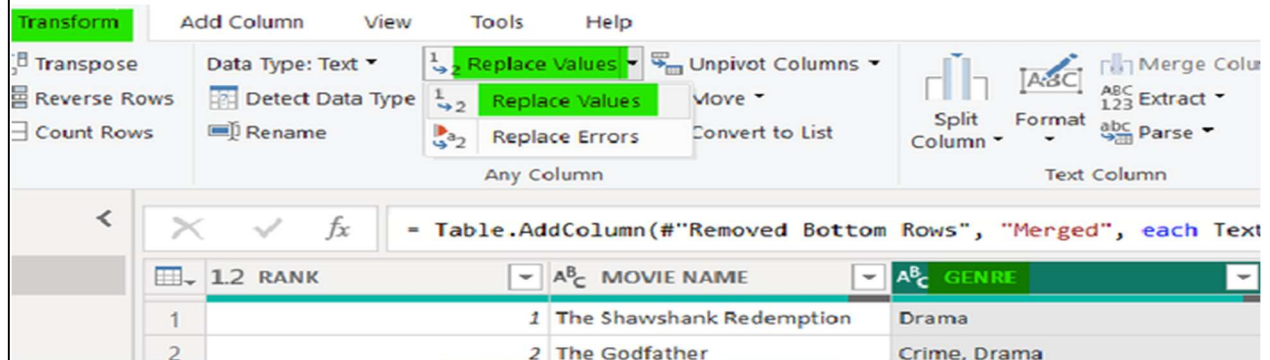
1.2 RANK	A ^B _C MOVIE NAME	A ^B _C GENRE	1.2 RATING	A ^B _C
1	1 The Shawshank Redemption	Drama	9.3	TH
2	2 The Godfather	Crime, Drama	9.2	TH

Output:

A ^B _C Merged
Drama ;) 9.3
Crime, Drama ;) 9.2
Biography, Drama, History
Crime, Drama, Mystery ;) 8.9
Comedy, Drama, Family ;) 8.6
Biography, Crime, Drama ;) 8.7
Drama, Romance ;) 6.7
9
Horror, Sci-Fi ;) 8.4
Animation, Adventure, Comedy ;) 8.3
Drama ;) 8.7
Crime, Drama
Action, Comedy, Crime ;) 6.8
Crime, Drama ;) 7.5
Horror, Mystery, Thriller ;) 8.5
Crime, Drama, Horror ;) 8.6

7. Replace Values Operation in Power BI

Replace values operation replaces some specific value to our desired value. It's present in the "Transform" bar as "Replace Values". Here, we have replaced "null" to "geeksforgeeks" for column "GENRE".



Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

geeksforgeeks

> Advanced options

OK

Output:



8. Split Column Operation in Power BI

- ❖ It's present in the "**Transform**" bar on the ribbon.
- ❖ In the transform bar lies an operation as "**split column**".
- ❖ We can split either by using a delimiter, by providing no. of characters or positions and so on.
- ❖ Here, we have split the column "**Merged**" using the customized delimiter " ;) " that splits the column into two columns "Merged.1" and "Merged.2".
The illustration is as follows :



Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

--Custom--
;)

Split at

- ☐ Left-most delimiter
☐ Right-most delimiter
☒ Each occurrence of the delimiter

> Advanced options

Quote Character

"

☐ Split using special characters

Insert special character



Before

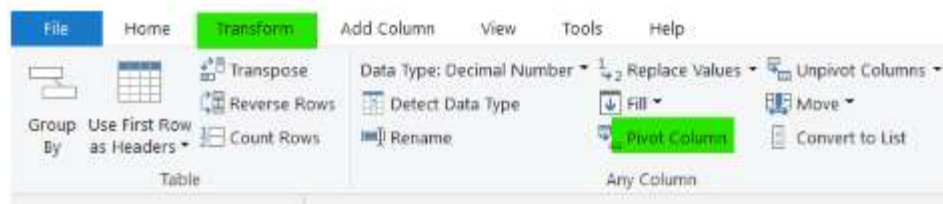
A ^B _C Merged
Drama ;) 9.3
Crime, Drama ;) 9.2
Biography, Drama, History
Crime, Drama, Mystery ;) 8.9
Comedy, Drama, Family ;) 8.6
Biography, Crime, Drama ;) 8.7
Drama, Romance ;) 6.7
9
Horror, Sci-Fi ;) 8.4
Animation, Adventure, Comedy ;) 8.3
Drama ;) 8.7
Crime, Drama
Action, Comedy, Crime ;) 6.8
Crime, Drama ;) 7.5
Horror, Mystery, Thriller ;) 8.5
Crime, Drama, Horror ;) 8.6

After

A ^B _C Merged.1	1.2 Merged.2
Drama	9.3
Crime, Drama	9.2
Biography, Drama, History	null
Crime, Drama, Mystery	8.9
Comedy, Drama, Family	8.6
Biography, Crime, Drama	8.7
Drama, Romance	6.7
9	null
Horror, Sci-Fi	8.4
Animation, Adventure, Comedy	8.3
Drama	8.7
Crime, Drama	null
Action, Comedy, Crime	6.8
Crime, Drama	7.5
Horror, Mystery, Thriller	8.5
Crime, Drama, Horror	8.6

9. Pivot Column Operation in Power BI

The pivot operation basically turns **rows into columns**. By default, the query editor does sum as aggregation which can also be set as don't aggregate or minimum or maximum or whatever as per the user's wish from available options. It is present in the "Transform" bar.



Data for pivot operation:

	1.2 Field 1	1.2 Field 2
1	1.1	2
2	2.6	9
3	4.3	2
4	2.6	3
5	3.9	2

Pivot Column

Use the names in column "Field 2" to create new columns.

Values Column ①

Field 1

Advanced options

Aggregate Value Function

Sum

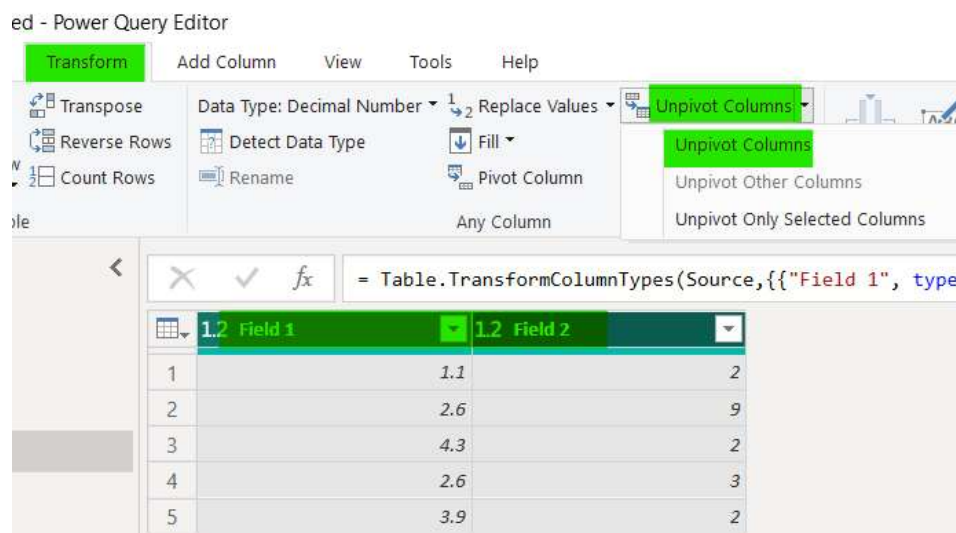
[Learn more about Pivot Column](#)

Output :

	1.2 2	1.2 9	1.2 3
1	9.3	2.6	2.6

10. Unpivot Column Operation in Power BI

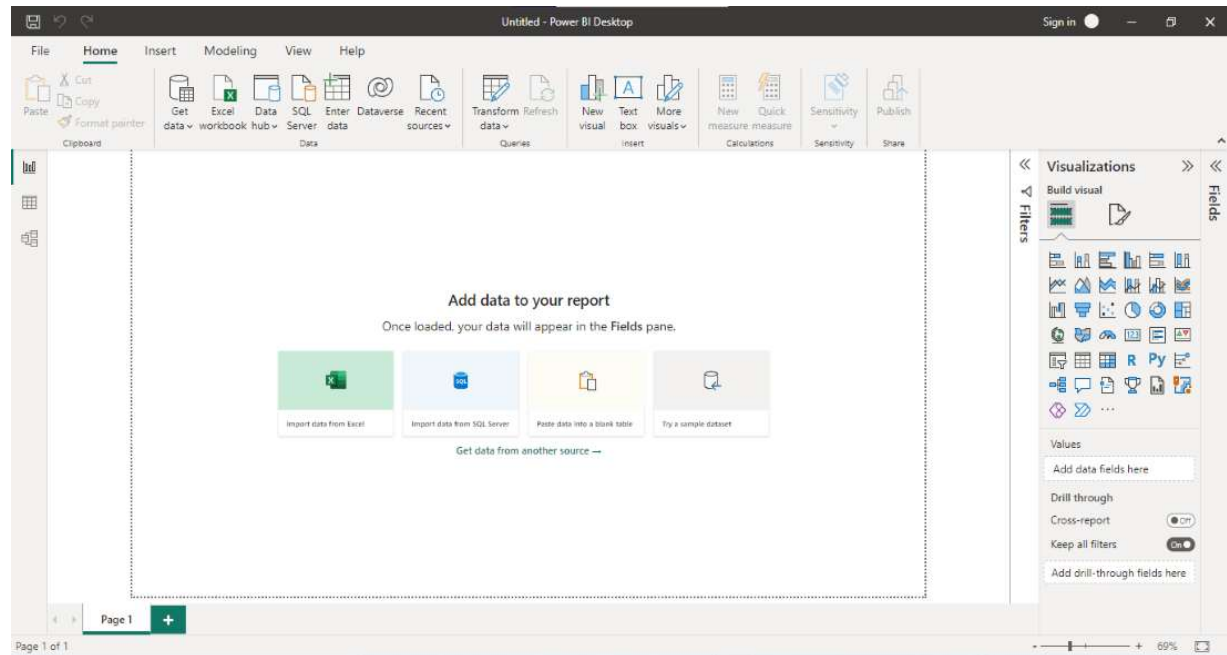
Unpivot column operation as the name also suggests does the opposite of what pivot does. Unpivot basically unpacks similar values and gathers them under one label. When we did unpivot on the same data we took for the pivot operation it produces the output:

Output:

	Attribute	Value
1	Field 1	1.1
2	Field 2	2
3	Field 1	2.6
4	Field 2	9
5	Field 1	4.3
6	Field 2	2
7	Field 1	2.6
8	Field 2	3
9	Field 1	3.9
10	Field 2	2

Power BI - How to edit in Power BI App?

- ❖ Power BI is a data visualization application that lets we connect, transform and find insights into the most pressing matters of business.
- ❖ It helps in sourcing your data and creating visual dashboards, KPIs, and reports by editing the data as per concerns.
- ❖ we can collaborate, share and integrate data easily across products documentations.
- ❖ It also has a wide range of visualizations, including charts, tables, and maps, and allows users to create interactive dashboards and reports.
- ❖ Power BI is widely used for data analysis, reporting, and business intelligence. With a cut-down risk of the misgovernment of data, including when data is exported out of your application, it proves worthy of use.



Use of Power BI

It is mainly used to create visualized reports and dashboards through which you can learn more about your data. Uploading report to the Power BI service enables users to interact and track the report being used. Here, we will go through the steps required for you to create and edit Reports and Dashboards.

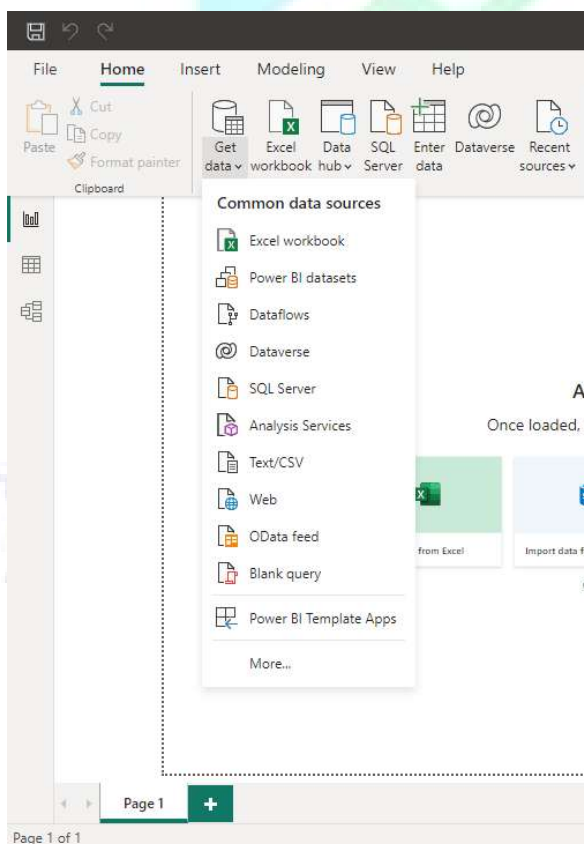
Basic Steps Required:

1. Extract the data
2. Transform the data as required and change the relationship between the tables using a power query.
3. Use DAX to perform the calculation on your data.
4. After getting your data ready, you can jump to your visualizations.
5. Add the graphs, charts, cards, etc., and create edits within them to make your report look more understandable.
6. Upload your dashboard to the cloud enabling your colleagues to access the dashboard and report.

Loading data

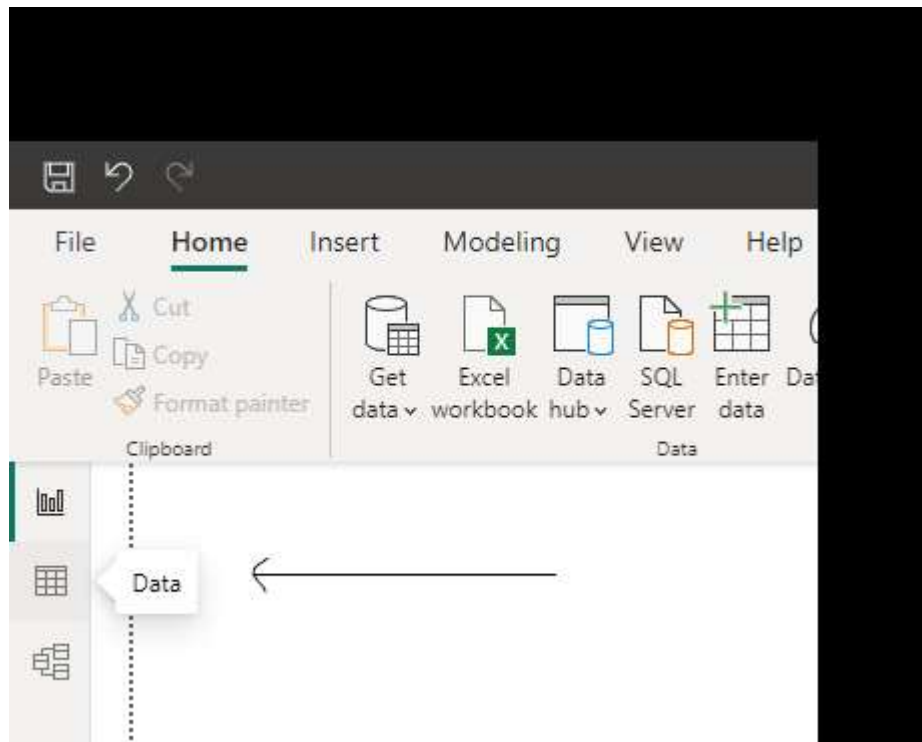
Let's start with how we can load data in application. we can load data from the 'Get data' option in the home tab. We can select from the following options:

- ❖ **Files:** Power BI can import statistics from Excel, CSV, and other record formats.
- ❖ **Databases:** It can connect to several databases, which include MySQL, Oracle, and SQL Server.
- ❖ **Direct Query:** It can retrieve the desired dataset by connecting to a data source without importing it beforehand.
- ❖ **Online Services:** It can connect to various online services for example Google Analytics, SalesForce, and SharePoint.
- ❖ **Live Connection:** An up-to-date live connection with a data source in real-time allows users to retrieve up-to-date data.

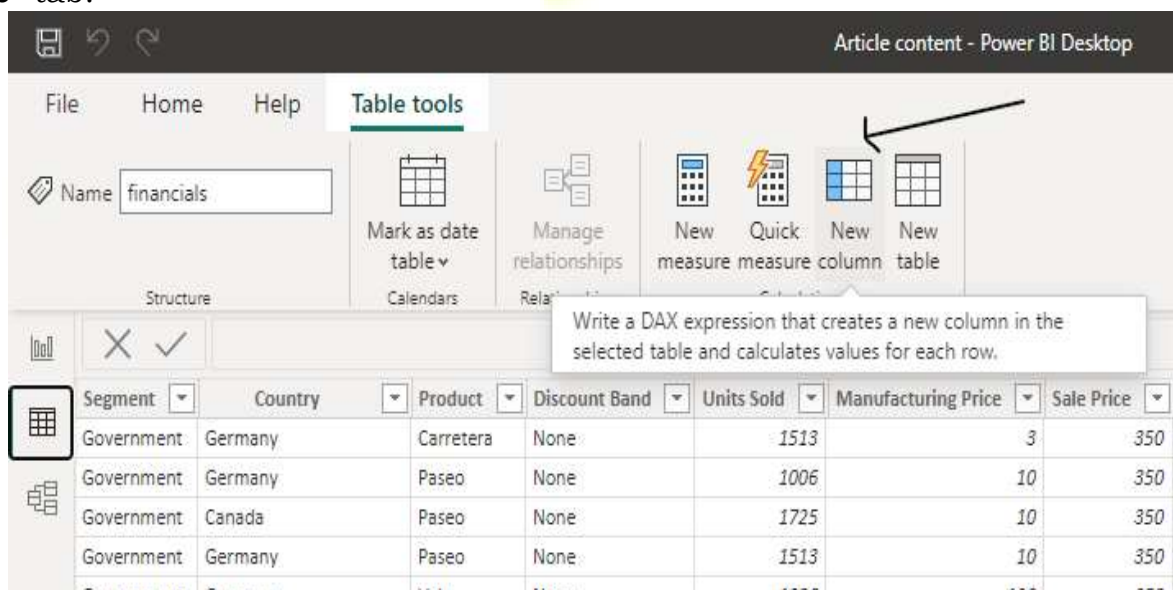


After the data gets loaded into the application, we can begin transforming and modeling data. We can select add new column option with custom options, and make own calculated field column.

Step 1: For this step Select "**Data**" from the three options on the left margin.



Step 2: Select "**New Measure**" from the "**Calculation**" section in the "**Table Tools**" tab.



Step 3: Now we can get desired functions with the table field's name and thus we'll be able to create a new column.

The top screenshot shows the 'Column tools' ribbon in Power BI Desktop. The formula bar contains '1 Column = SQRT(' and a dropdown menu of functions is open. The functions listed are: FACT, FALSE, FILTER, FILTERS, financials, financials[Sales], financials[COGS], financials[Country], financials[Date], financials[Discount Band], financials[Discounts], and financials[Gross Sales]. The bottom screenshot shows the 'Measure tools' ribbon. The formula bar contains '1 Measure = S' and a dropdown menu of measures is open. The measures listed are: ISSELECTEDMEASURE, ISSUBTOTAL, NONVISUAL, SELECTEDMEASURE, SELECTEDMEASUREFORMATSTRING, SELECTEDMEASURENAME, SUBSTITUTE, SUBSTITUTEWITHINDEX, SUM, SUMMARIZE, SUMMARIZECOLUMNS, and SUMX. A small table is visible in the bottom right corner of the bottom screenshot.

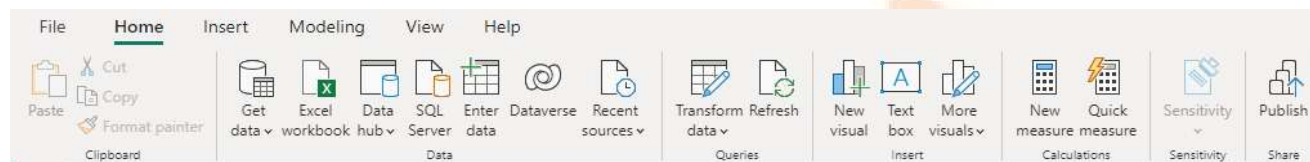
Sum of Sale Price	Year	Month	Sum
4145	2013	September	71
8290	2013	October	1,61
4145	2013	November	71
20725			3,87

Step 4: After this step, we'll complete required columns or measures. We should use the "New Column" method when we require this column to be present at data refreshes as well because it gets defined in the table. Whereas measure works as a filter on table and it is faster because they are not stored in the memory. Power BI has a built-in Power Query Editor that makes data visuals-ready. It allows to make advanced changes to clean data for perfect visualizations and get model done.

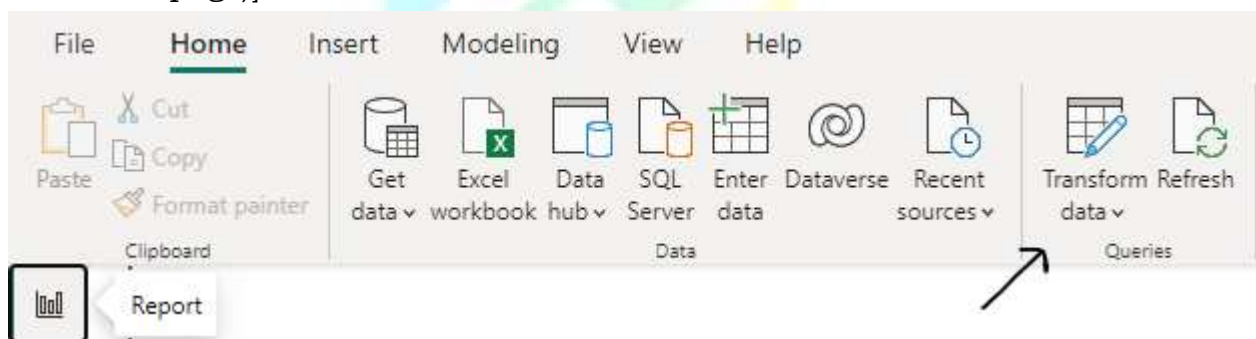
Power Query

Power Query is used to transform data by performing tasks such as filtering, grouping, and pivoting. It helps to create calculated columns and measures by using the formula bar and functions. After the cleaning and transformation of your data, you can move on to visualizations. Following are the steps to use Power Query.

Step 1: Select "Home" from the top menu.

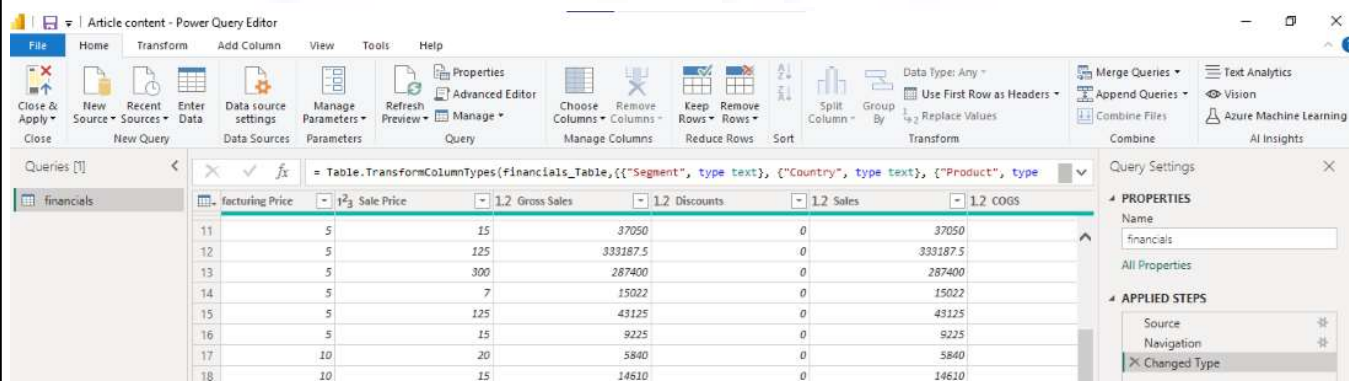


Step 2: From the queries section select **"Transform data"**. (A new window will open with Power Query Editor), (If no data connection is made yet then it'd show a blank page).

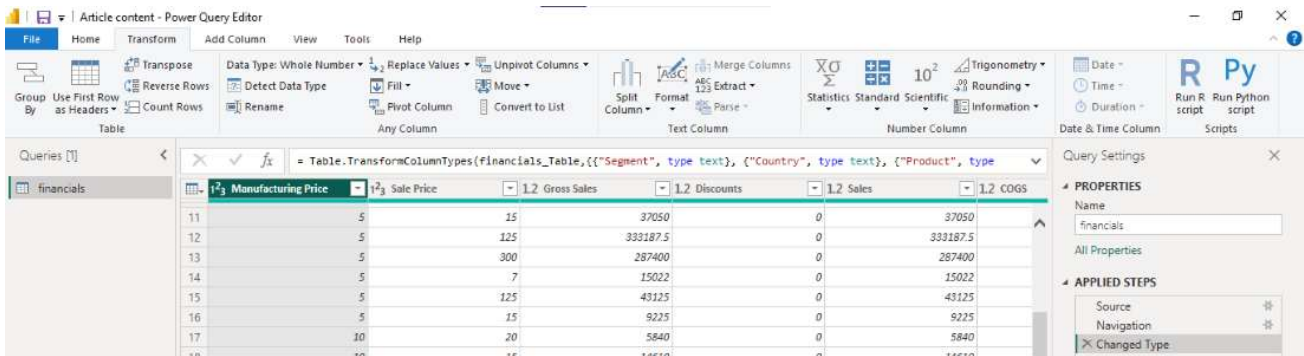


Step 3: On this page we'll get to add, change, view, transform, and correct data.

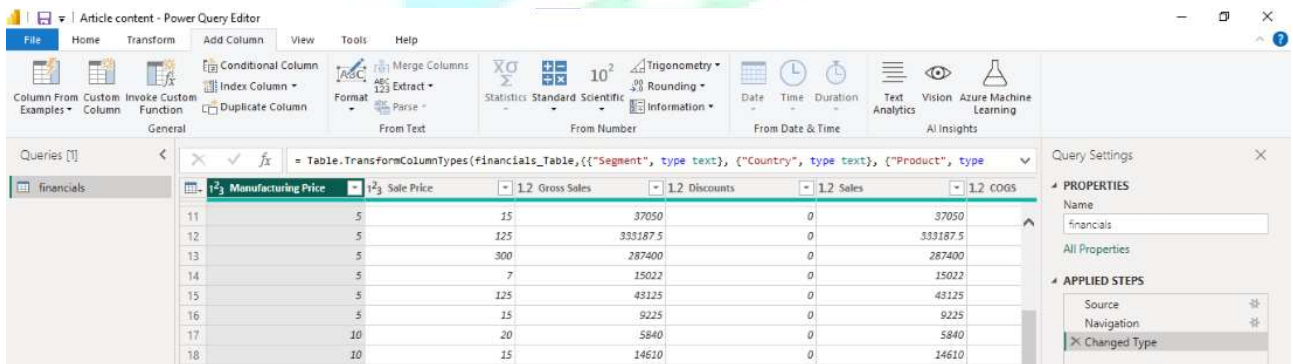
❖ The "Home" tab provides data source options.



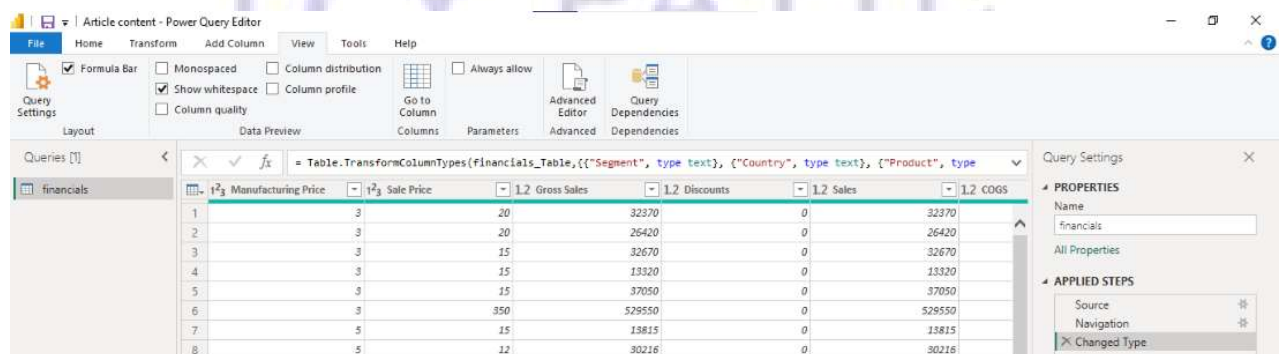
- ❖ The "**Transform**" tab includes addition, deletion, splitting, and changing data types functions.



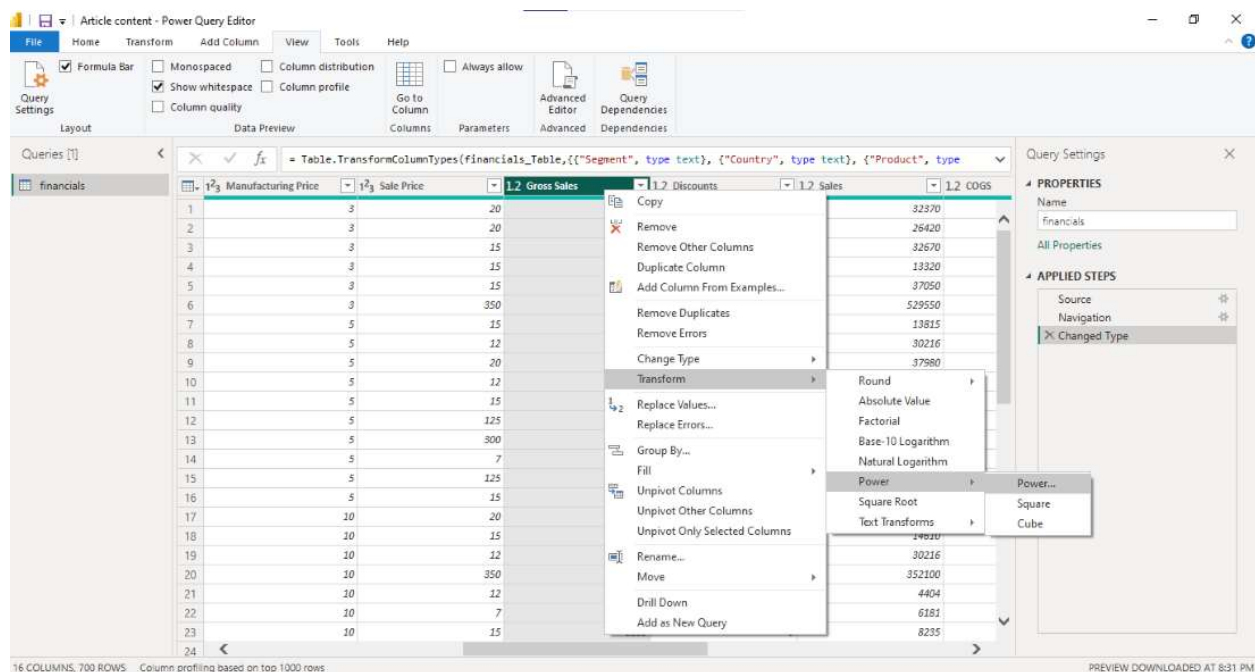
- ❖ The "**Add Column**" tab provides the column addition and formatting of the same.



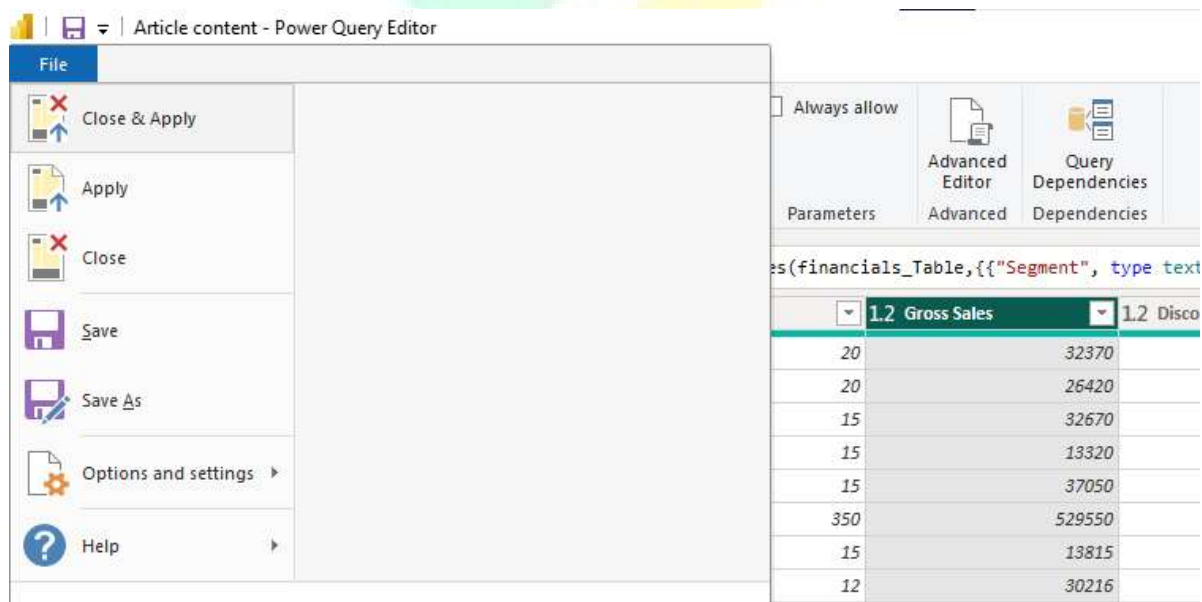
- ❖ The "**View**" tab provides how the page is displayed.



Step 4: With a right-click on your column you can make different changes



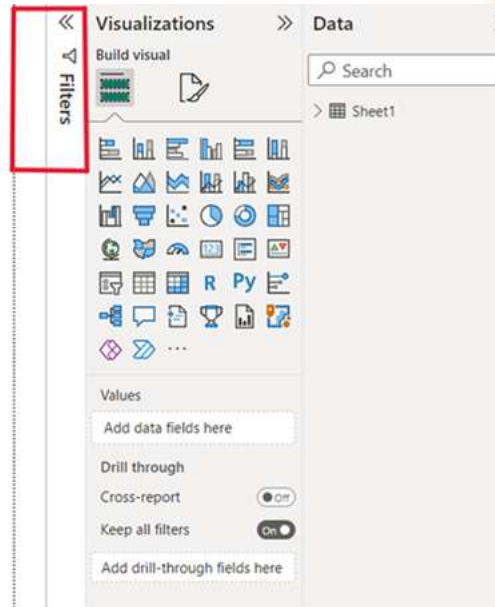
Step 5: After making the necessary transformations, click on **"Close & Apply"** by clicking on the **"File"** tab to save the changes and return to the Power BI Desktop.



Now data is now ready to be used in visualizations and reports.

PBI Filters

- ❖ Filters in Power BI play a important role in shaping meaningful reports by narrowing down data to only what matters.
- ❖ They help you focus on relevant subsets of data, ensuring visuals and dashboards are targeted, clear and actionable.
- ❖ Whether applying filters at the visual, page or report level, these tools provide a systematic and consistent way to refine analysis.



- ❖ Filters in Power BI allow you to display only the data we need based on defined criteria.
- ❖ We can apply them in various ways to streamline the reports and dashboards, enabling data-driven decision-making with ease.

Types of Power BI Filters

Power BI provides four main types of filters that can be applied at different levels which helps us to manage our data more effectively:

Visual-level Filters:

- ❖ These filters apply to individual visuals within a report like tables, charts or cards.
- ❖ They allow us to control how data is presented within a specific visual, giving us flexibility to focus on the data that's most important for that visual.

- ❖ We can use sub-filter modes like Basic filters, Advanced filters and Top-N filters to adjust the data within that visual.

Page-level Filters:

- ❖ Page-level filters apply to all visuals on a specific page of the report.
- ❖ When a page-level filter is applied, it automatically updates every visual on the page based on the filter's criteria.
- ❖ This is useful for ensuring that the entire page reflects the same filtered data helps in providing a cohesive view of the report.

Drill-through Filters:

- ❖ Drill-through filters allow us to explore data in more depth by navigating between different pages.
- ❖ We can set up drill-through filters to link pages and filter the data dynamically so we can analyze detailed subsets of data without overwhelming the user with too much information on a single page.

Report-level Filters:

- ❖ These filters apply to all visuals across all pages in the report.
- ❖ Report-level filters are useful when we want to maintain consistency across all pages helps in ensuring that the same filtering criteria are applied throughout the report.

Add a Filter to a Report in Power BI

There are three main filter modes in Power BI that help us control which data is displayed:

1. Basic Filtering

- ❖ Basic filtering is the simplest and most intuitive filter mode.
- ❖ It provides a scrollable and searchable list of values for a specific field in data.
- ❖ We can quickly search for a value and select it or use checkboxes to select multiple values.

Country
is (All)

Filter type ⓘ

Basic filtering ▼

Search

☐	Select all	
☐	(Blank)	1
☐	Canada	12
☐	France	9
☐	Germany	15
☐	Mexico	8
☐	United States of America	5

- ❖ This mode is ideal for situations where we simply want to focus on one or more data points without needing to apply complex conditions.

2. Advance Filtering

- ❖ Advanced filtering gives us greater control over how data is filtered by allowing us to define more specific criteria.
- ❖ Instead of just selecting from a list, we can apply rules such as greater than, less than, between a range of values or even exclude certain values.

Filters on this visual ...

Country
does not contain 'Canada' and does not start with 'Germany'

Filter type ⓘ

Advanced filtering ▼

Show items when the value

does not contain ▼

Canada

☒ And ☐ Or

does not start with ▼

Germany

Apply filter

- ❖ This mode is useful when we need to filter data based on numerical ranges or specific conditions.

3. Top-N Filtering

- ❖ Top-N filtering is used when we want to focus on the highest or lowest N values in a dataset.
- ❖ This filter type allows us to quickly identify top performers or outliers.
- ❖ We can specify how many top or bottom items we want to display such as the top 5 countries by sales or the bottom 3 products by revenue.

Filters on this visual

Country
top 2 by First Country

Filter type ⓘ
Top N

Show items
Top 2

By value
First Country

Apply filter

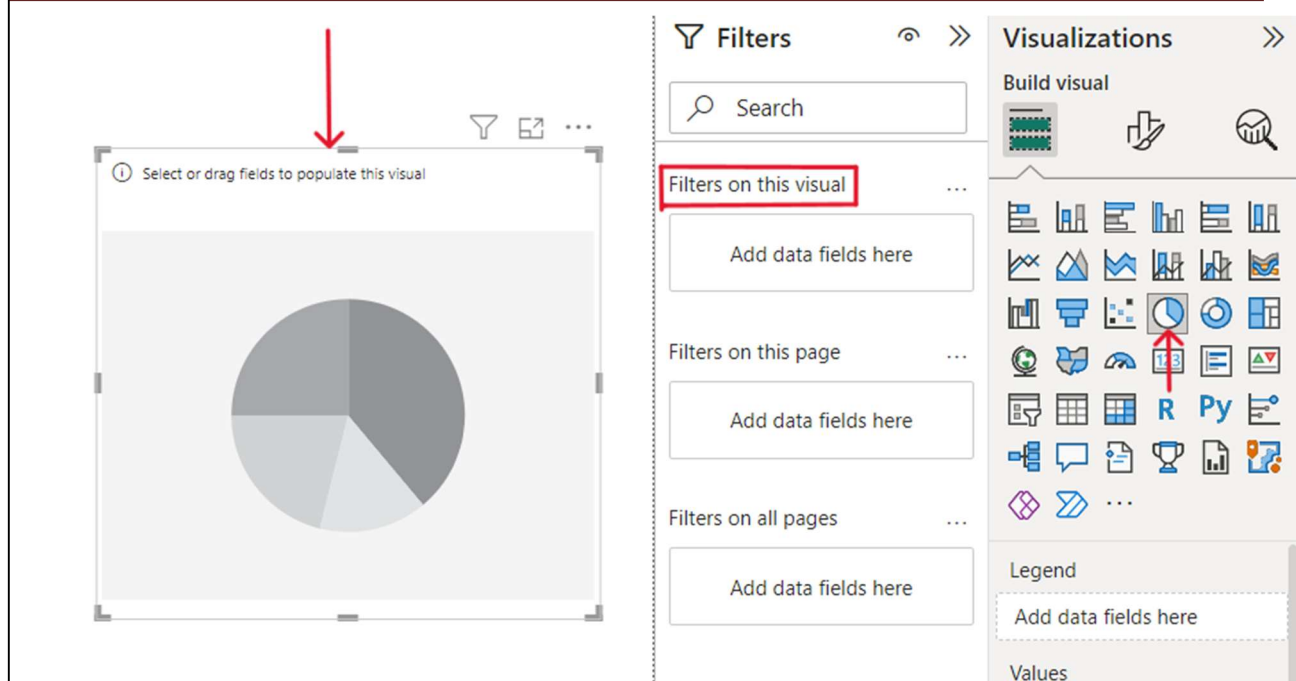
- ❖ Top-N filters are useful in identifying trends or analyzing the extremes within our data.

How to add Power BI Filter to a Visual?

Step 1: Select the Visual

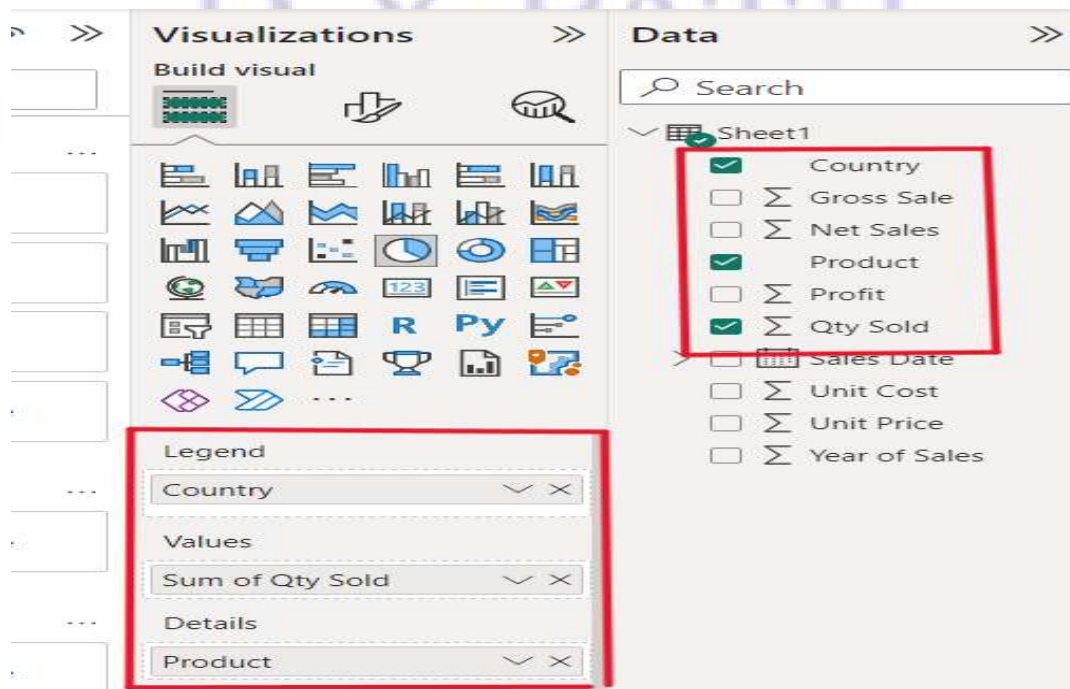
First, we select the visual we want to apply the filter to. This could be any chart, table or card.

- ❖ Once we select the visual the "**Filters on this visual**" option will appear in the **Filters** panel on the right side of the screen.
- ❖ For example, after selecting a pie chart, we will see the "**Filters on this visual**" section appear showing that we can now apply filters specific to that visual.



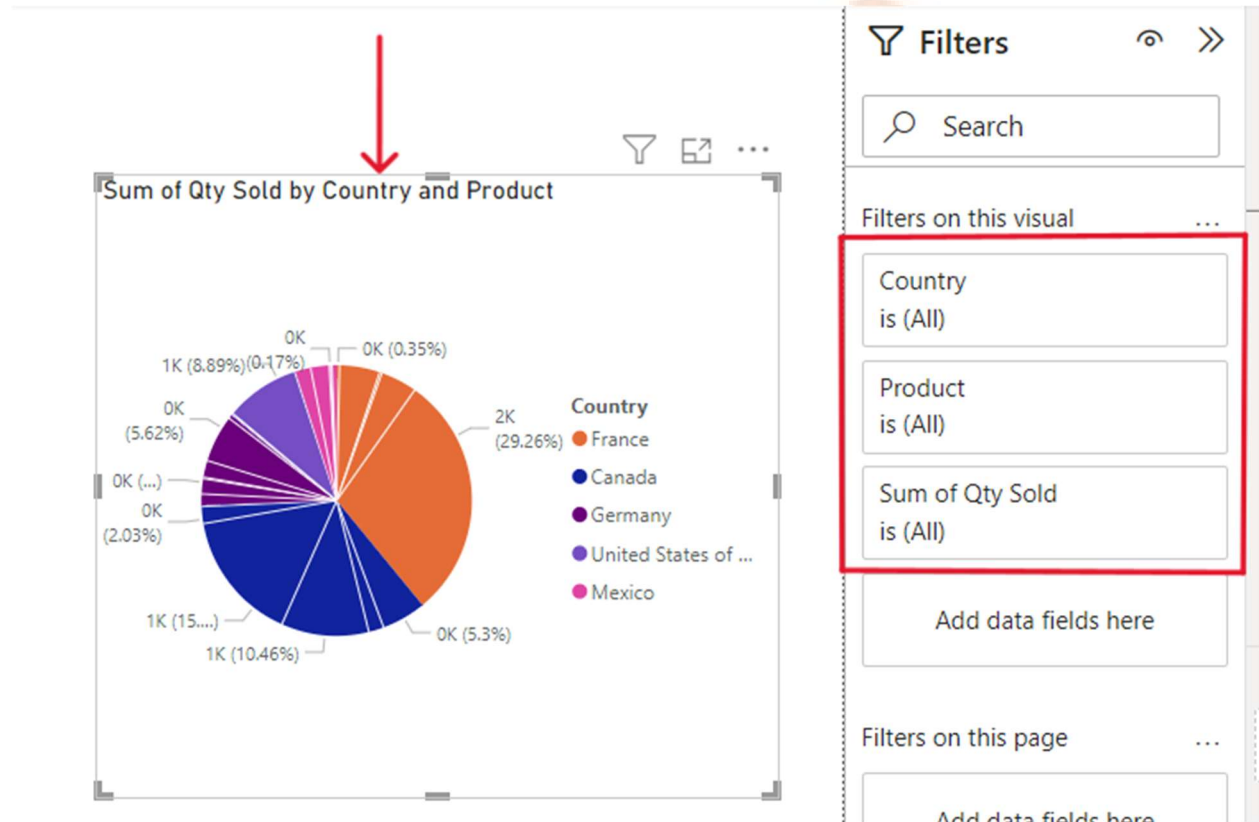
Step 2: Drag and Drop Fields

- ❖ Next, we drag and drop the desired fields from the **Fields** pane into the appropriate filter areas in the Filters panel.
- ❖ For example, we can drag the **Country**, **Product** and **Quantity Sold** fields from the data panel and drop them into the relevant filter sections.
- ❖ This action will populate the visual with data based on these fields.



Step 3: Apply the Filter

- ❖ After adding the fields, the visual will automatically update to reflect the applied filter.
- ❖ In the case of a pie chart, as seen in the screenshot, the visual will display the filtered data accordingly.
- ❖ Filters are typically applied automatically when we add fields but we can fine-tune the filter options to control which data is displayed more precisely.



How to add a Power BI Filter to a Page?

Step 1: Create Multiple Visuals

First, create at least two visuals on the page like a pie chart and a table. This helps us see the effect of page-level filters on multiple visuals.

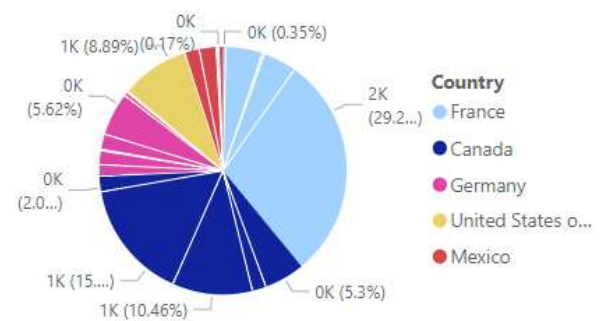
The screenshot shows a data visualization tool interface. On the left, there are two visual placeholders, each with a red arrow pointing to it and the instruction "Select or drag fields to populate this visual". The top placeholder is a table, and the bottom one is a pie chart. On the right, there is a sidebar with two main sections: "Filters" and "Visualizations". The "Filters" section has a search bar and two sections for adding data fields: "Filters on this page" and "Filters on all pages". The "Visualizations" section has a "Build visual" button and a grid of visualization icons. A red arrow points to the "Table" icon in the grid. Below the grid, there are sections for "Values", "Drill through", "Cross-report", and "Keep all filters".

Step 2: Use Visual-Level Filters

- ❖ Initially, we apply visual-level filters to populate each visual individually.
- ❖ As we can see in the screenshot we populate the visuals (pie chart and table) by selecting the relevant fields like Product, Sales, Quantity, etc from the data panel.

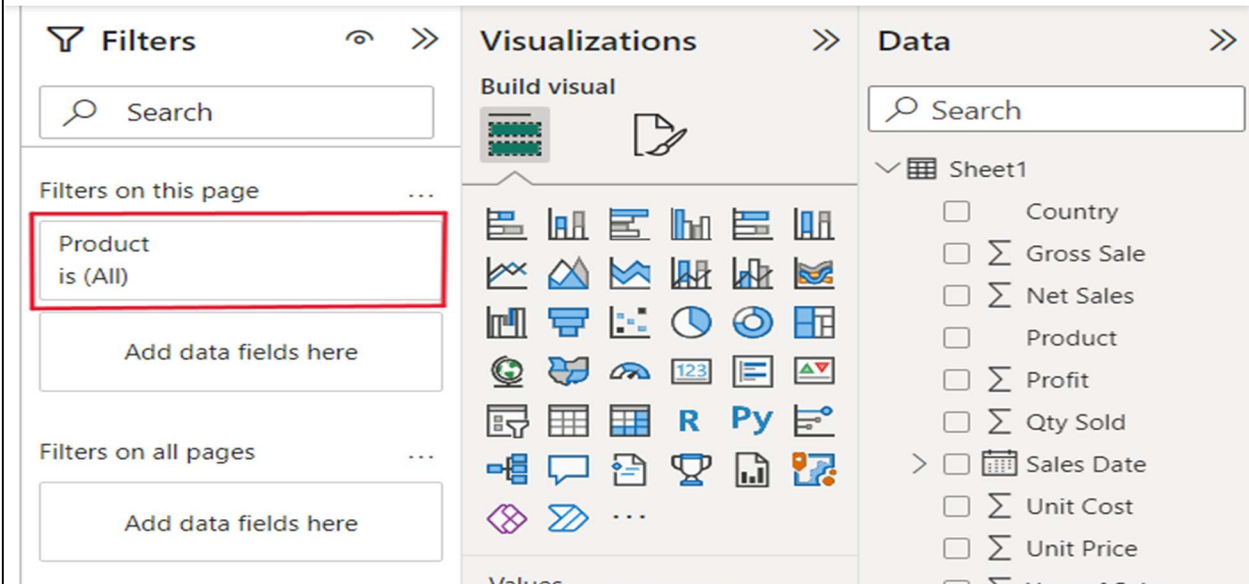
Country	Product	Sum of Qty Sold
Canada	CPU	365
Canada	Hard Disk	125
Canada	Keyboard	720
Canada	Pad	1075
Canada	Pen	140
Canada	Pencil	12
France	CPU	24
France	Hard Disk	330
France	Keyboard	20
France	Mouse	300
France	Pen	2014
Germany	CPU	90
Germany	Hard Disk	125
Germany	Keyboard	12
Germany	Mouse	137
Germany	Pen	287
Total		6882

Sum of Qty Sold by Country and Product

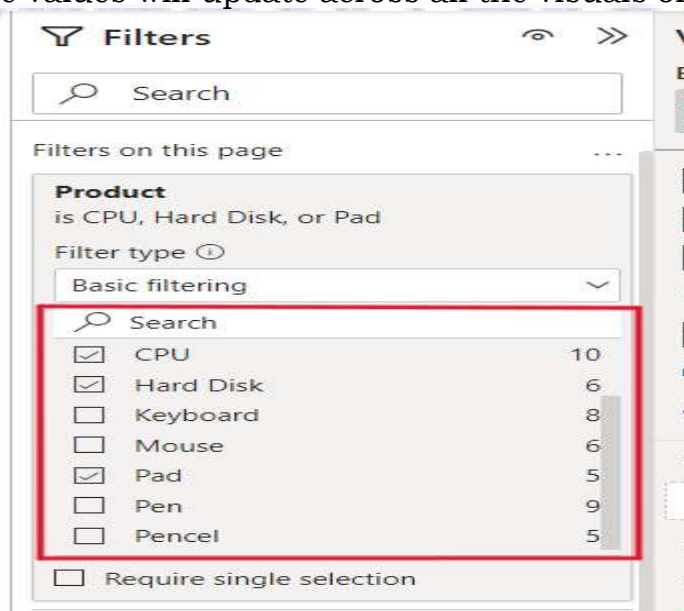


Step 3: Apply Page-Level Filters

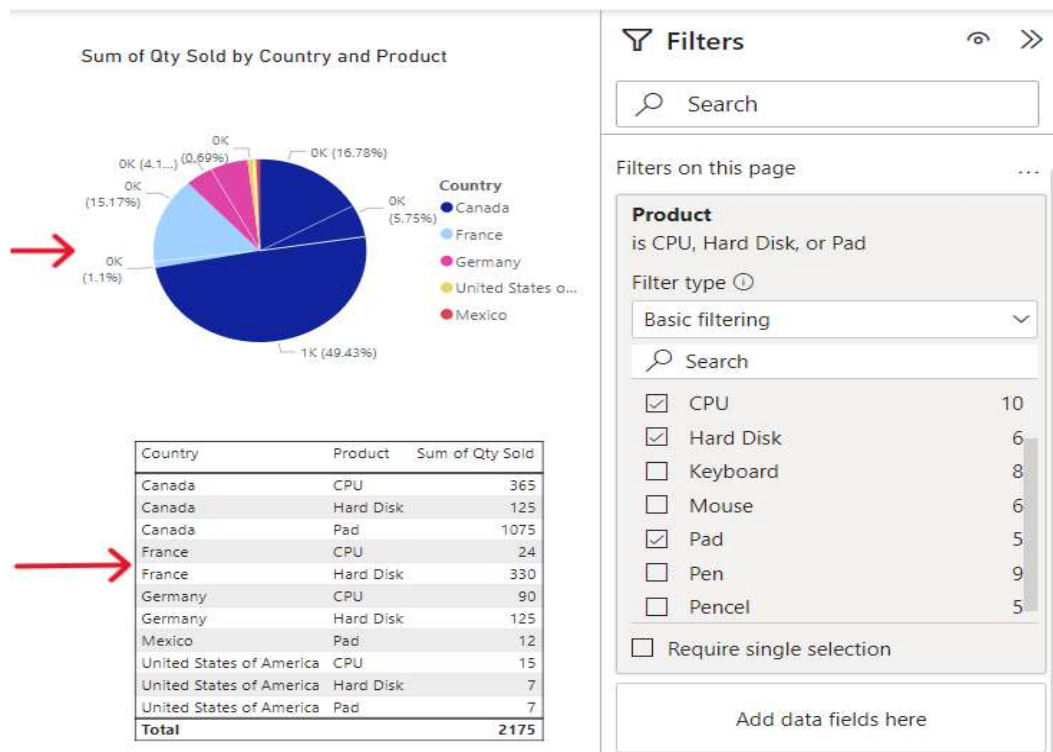
- ❖ To apply a page-level filter, we drag and drop the desired data fields into the "Filters on this page" section.
- ❖ For example, we drop the Product field into this section. This will allow us to filter the entire page based on the selected field.

**Step 4: Select Filter Values**

- ❖ Next, we select the specific values we want to filter across all visuals on the page.
- ❖ For example, we select CPU, Hard Disk and Pad from the Product field and these values will update across all the visuals on the page.



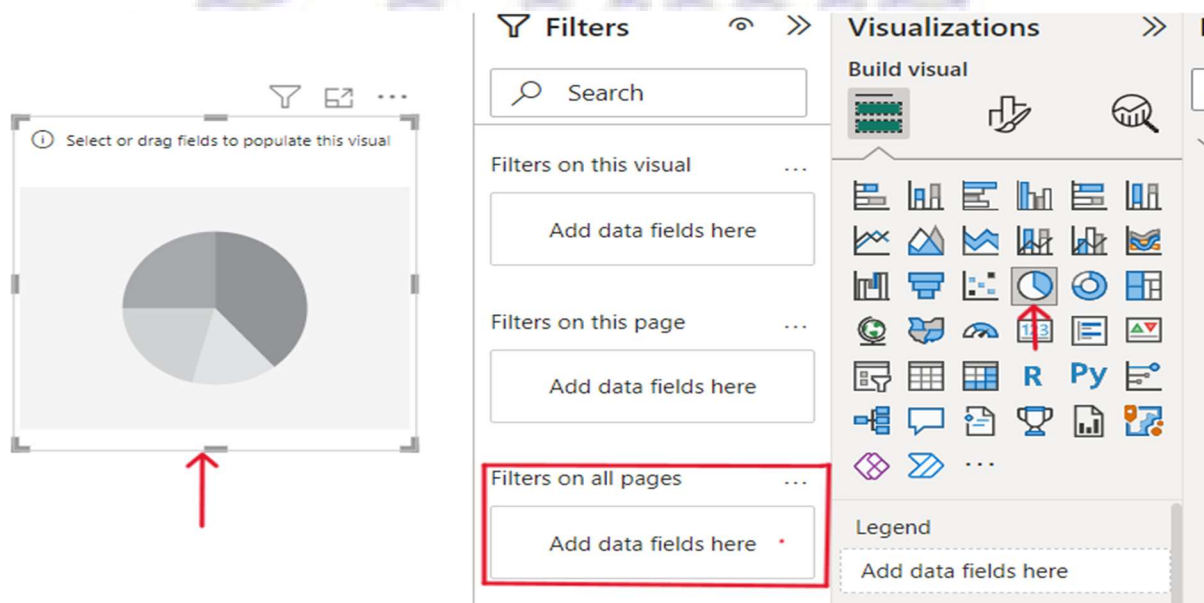
The pie chart and table two visuals get updated by in its product field.



How to add Power BI Filter to a Report?

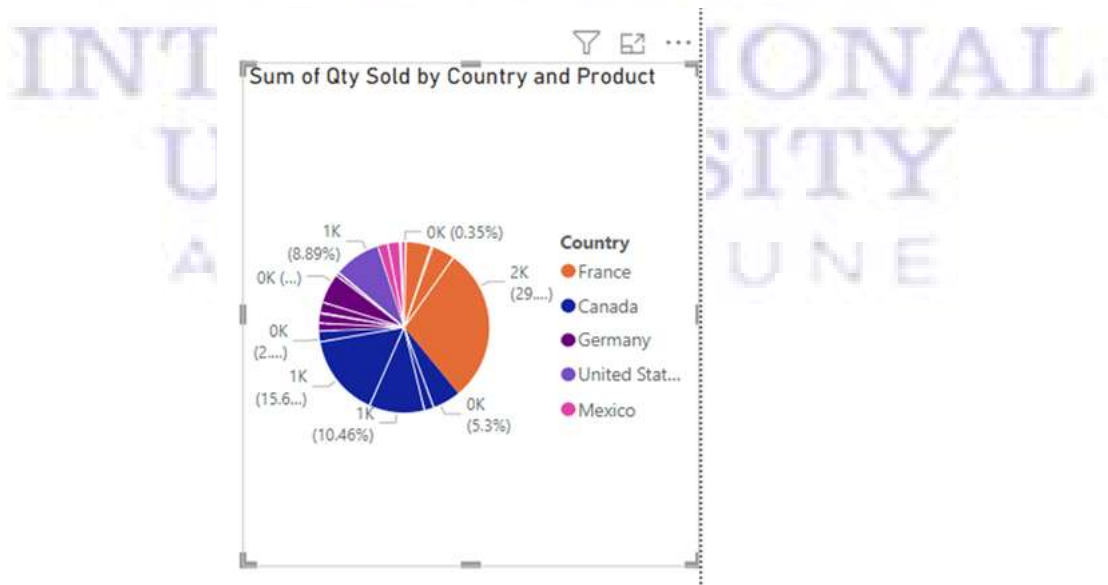
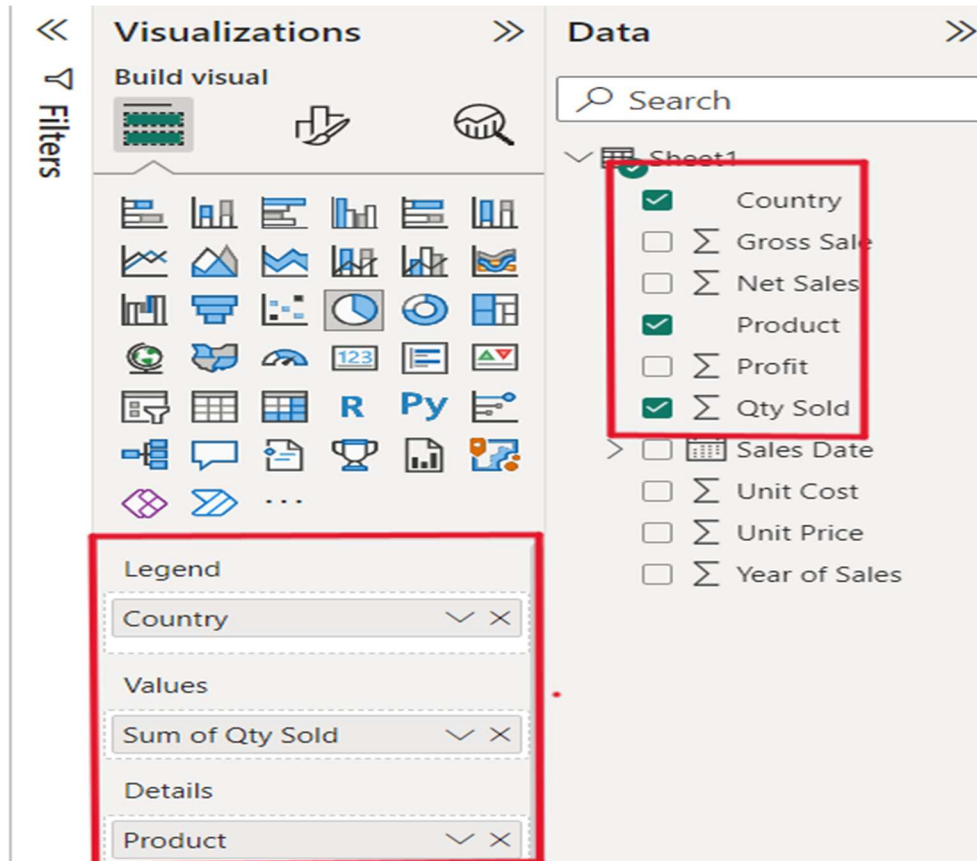
Step 1: Create a Visual

To use the report level filter create a visual on the page. As we can see in the below screenshot we created the visual that is a pie chart.



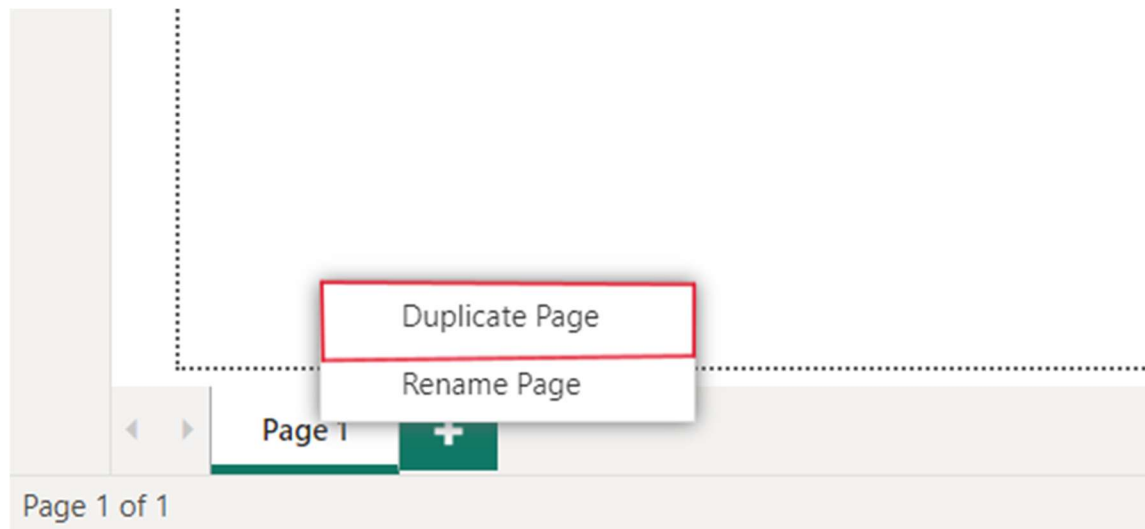
Step 2: Populate the Visual

Drag the relevant data fields from the Fields panel and drop them into the appropriate areas in the visual. Once this is done, the pie chart will display the data accordingly.



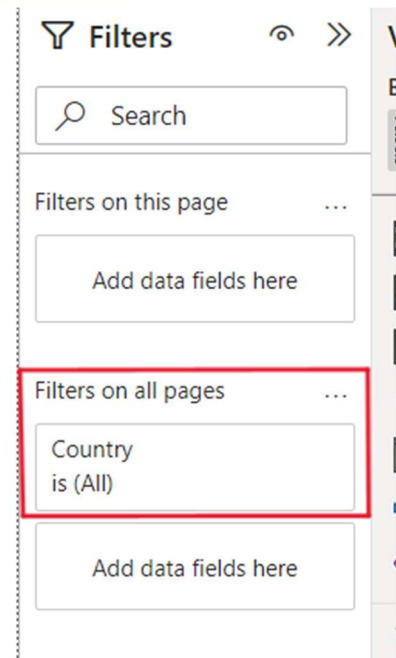
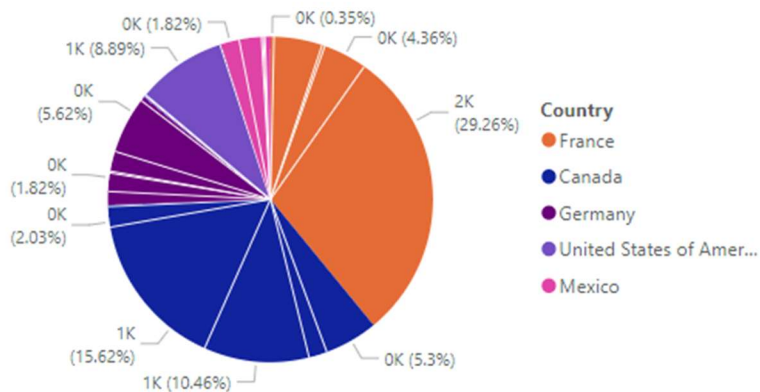
Step 3: Duplicate the Page

Since we need at least two pages to apply a report-level filter, right-click on Page 1 and select Duplicate Page. This creates a duplicate page with the same visual setup.

**Step 4: Apply the Report-Level Filter**

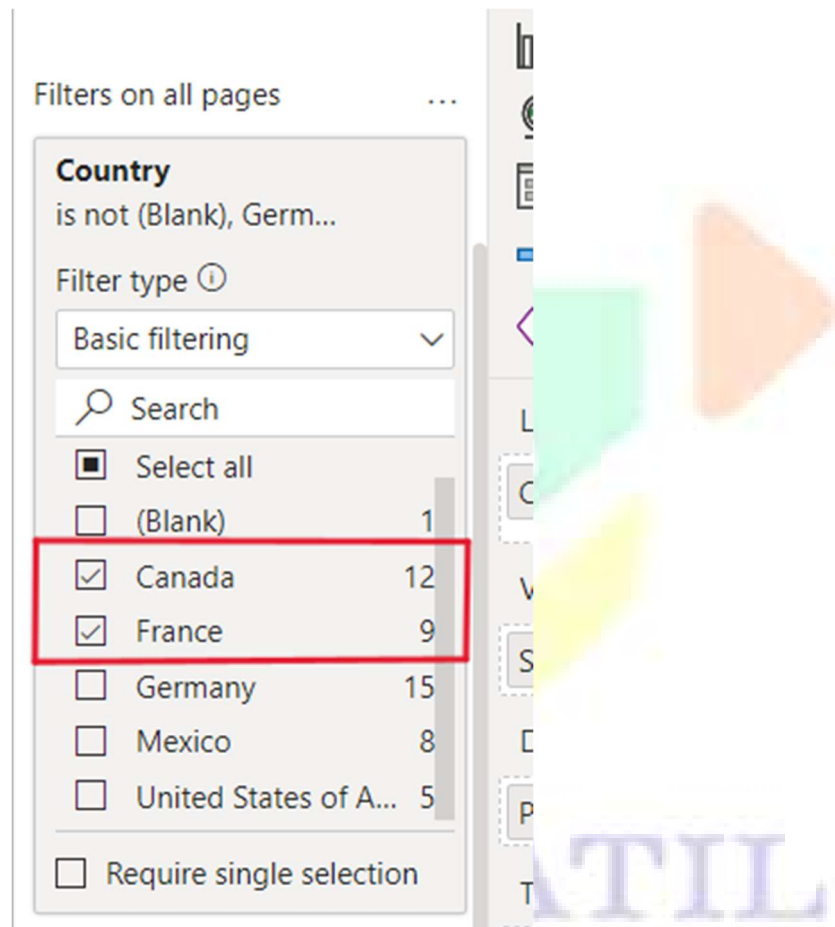
Drag the desired data field like Country into the "Filters on all Pages" section. This applies the filter to all visuals across the report.

Sum of Qty Sold by Country and Product

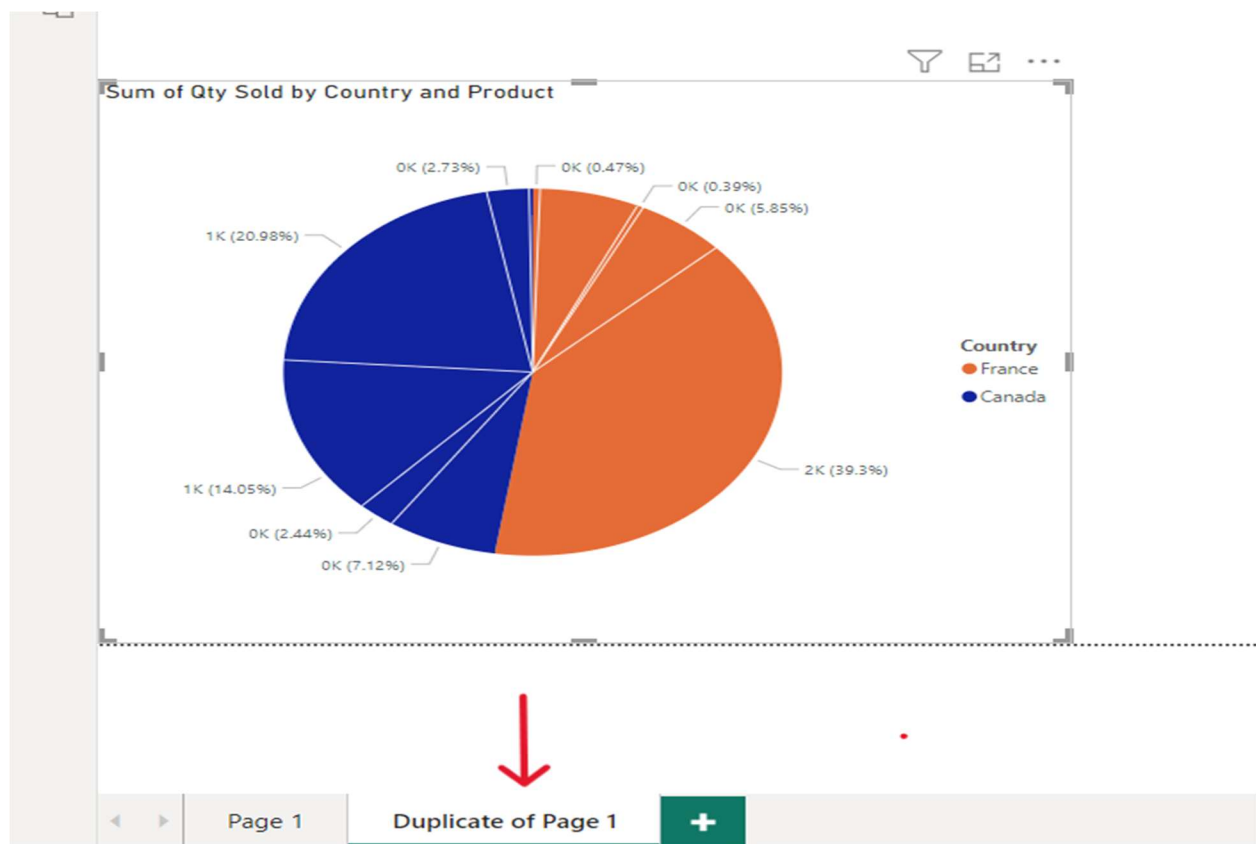


Step 5:

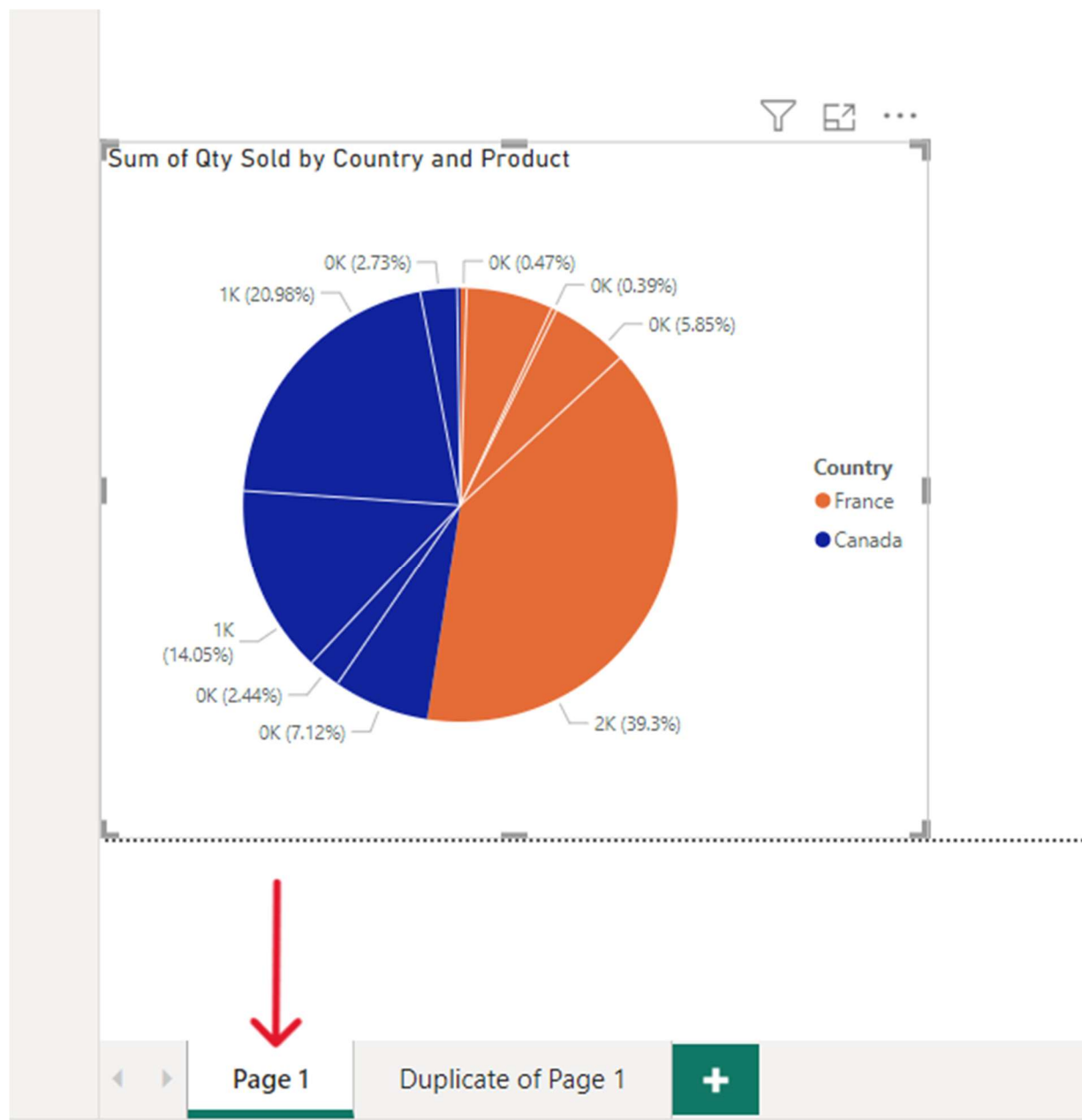
Choose the specific values from the data field such as Canada and France. These values will apply the filter across the entire report.

**Step 6:** View the Filtered Report

After selecting the filter values, the filter will be applied to the entire report including all pages. The same filter criteria will automatically apply to Page 1 from the duplicate page.



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Column & Row Transformations

- ❖ Pivot: Rows to Columns.
- ❖ Unpivot: Column to rows.
- ❖ Transpose: Rows to Columns and Column to Rows.

Pivot: Pivoting is used to convert rows into columns based on unique values in a column.

Step-by-Step Pivot Process

1. **Launch Power Query:** From the Home tab in Power BI Desktop, click Transform Data to open the editor.
2. **Select the Attribute Column:** Click the header of the column containing the values you want to turn into new column headers.
3. **Initiate Pivot:** Navigate to the Transform tab on the top ribbon and select Pivot Column.
4. **Configure Values:** In the dialog box that appears:
 - ❖ **Values Column:** Select the column that contains the actual data you want to populate under the new headers.
 - ❖ **Advanced Options (Aggregation):** Click the "Advanced options" dropdown to choose how to summarize the data. The default is Sum, but you can also choose Count, Average, Min, Max, or Don't Aggregate.
5. **Finalise:** Click OK. Power BI will reshape the table, and we can then click Close & Apply on the Home tab to save the changes to data model.

☒ Example Scenario

We have this data:

Student	Subject	Marks
A	Math	80
A	Science	75
B	Math	85
B	Science	78

We want this output:

Student	Math	Science
A	80	75
B	85	78

UnPivot: Pivoting is used to convert columns into rows based on unique values in a row.

Steps to Unpivot Columns

1. **Open Power Query:** In Power BI Desktop, go to the Home tab and click Transform Data.
2. **Select Target Columns:**
 - ❖ To unpivot specific columns: Hold Ctrl and select the columns you want to transform into rows.
 - ❖ To unpivot everything else: Select only the "fixed" columns (like ID or Category) that should stay exactly as they are.

3. Apply Transformation: Navigate to the Transform tab and click the Unpivot Columns dropdown to choose one of these three methods:

- ❖ Unpivot Columns: Best when you have a fixed, known set of columns.
- ❖ Unpivot Other Columns: Recommended for dynamic data. If you add a new column (e.g., a new month) to your source later, Power BI will automatically unpivot it during refresh.
- ❖ Unpivot Only Selected Columns: Only transforms the columns you currently have highlighted.

4. Rename Resulting Columns: Power BI will create two new columns: Attribute (the old headers) and Value (the data points). Double-click the headers to rename them to something meaningful, like "Month" and "Sales".

5. Finalise: Click Close & Apply on the Home tab.

☒ Example Scenario

Before Unpivot (Pivoted Data)

Student Math Science

A	80	75
B	85	78

After Unpivot (Desired Output)

Student Subject Marks

A	Math	80
A	Science	75
B	Math	85
B	Science	78

Transpose function is a "global" switch: it flips the entire table's orientation so that every single row becomes a column and vice versa.

The Step-by-Step Flow

- ❖ **Demote Headers:** If your headers are data points that need to be flipped, go to the Transform tab and select Use Headers as First Row.
- ❖ **Transpose:** Click Transpose in the Transform ribbon. Your vertical lists are now horizontal.
- ❖ **Promote Headers:** Once the flip is done, your new headers are likely sitting in the first row. Click Use First Row as Headers to move them up.
- ❖ **Set Data Types:** Transposing often clears Data Types. Select your columns and use Detect Data Type to ensure numbers and dates are recognized correctly.

☒ Example

Before Transpose

Name	Age	City
A	25	Pune
B	30	Mumbai

After Transpose

	Column1	Column2
Name	A	B
Age	25	30
City	Pune	Mumbai

Important Difference

Feature	Pivot	Unpivot	Transpose
Rows → Columns	☒	☒	☒ (entire table)
Columns → Rows	☒	☒	☒ (entire table)
Based on category column	Yes	Yes	No

Combine Queries

Combine Queries is used to merge or append multiple tables into one table in Power Query Editor.

Merge Queries

Merge Queries combine data from two tables based on a common column, adding related information from one table to another in a horizontal manner.

- ❖ Works horizontally by adding new columns
- ❖ Matches rows using a common key column
- ❖ Default join type is Left Outer Join
- ❖ Keeps all rows from the primary table
- ❖ Adds matching data from the secondary table
- ❖ Row count usually remains unchanged unless another join type is used

Consider the following two tables:

1.Sales Dataset

Column1	Column2	Column3	Column4	Column5	Column6	Column7
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
order_id	product_id	store_id	order_date	order_date_2	sales	revenue
40911	P0392	S0001	1/1/2019	1-Jan-19	0	0
40912	P0018	S0071	1/1/2019	1-Jan-19	0	0
40913	P0565	S0096	1/1/2019	1-Jan-19	0	0
40914	P0372	S0121	1/1/2019	1-Jan-19	0	0
40915	P0543	S0101	1/1/2019	1-Jan-19	2	7.41
40916	P0286	S0046	1/1/2019	1-Jan-19	0	0
40917	P0428	S0085	1/1/2019	1-Jan-19	3	83.19
40918	P0171	S0079	1/1/2019	1-Jan-19	0	0
40919	P0125	S0084	1/1/2019	1-Jan-19	0	0
40920	P0500	S0103	1/1/2019	1-Jan-19	0	0
40921	P0399	S0125	1/1/2019	1-Jan-19	0	0
40922	P0499	S0093	1/1/2019	1-Jan-19	0	0

2. Product Dataset

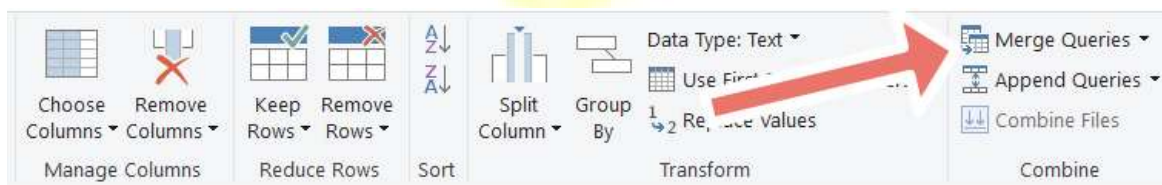
A ^B _C product_id	A ^B _C product (brand)	A ^B _C type	1.2 length	t ² ₃ depth
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 98% Error 0% Empty 2%	Valid 98% Error 0% Empty 2%
P0000	Serum (Livon)	Indian & Exotic Herbs	1.968505	20
P0001	Hand Wash - Moisture Shield (Savlon)	Hair Oil & Serum	5.3149635	22
P0002	Good Day Butter Cookies (Britannia)	Hand Wash & Sanitizers	8.661422	40
P0004	Happy Happy Choco-Chip Cookies (Parle)	Cookies	0.787402	13
P0005	50-50 Timepass Salted Biscuits (Britannia)	Glucose & Milk Biscuits	6.299216	30
P0006	Tiger Elaichi Cream Biscuits (Britannia)	Salted Biscuits	3.3464585	15
P0007	Bounce Biscuits - Choco Creme (Sunfeast)	Glucose & Milk Biscuits	0.787402	22
P0008	50-50 Timepass Biscuits (Britannia)	Cream Biscuits & Wafers	1.968505	16
P0009	Tiger Chocolate Cream Biscuits (Britannia)	Salted Biscuits	1.968505	18
P0010	Biscuits - Magix Krems Choc (Parle)	Cream Biscuits & Wafers	0.787402	22
P0011	Dreams Cup Cake - Choco (Elite)	Cream Biscuits & Wafers	3.543309	22
P0012	Layer Cake - Chocolate (Winkies)	Muffins & Cup Cakes	3.93701	20
P0013	Layer Cake - Orange (Winkies)	Tea Cakes & Slice Cakes	2.7952771	28
P0014	Sugar Coated Chocolate (Cadbury Gems)	Tea Cakes & Slice Cakes	12.7952825	6

Steps to Merge Queries

Step 1: Open Power Query Editor and select the primary table you want to merge from the left pane.

Step 2: With the table selected go to the Home tab to access the Ribbon menu.

Step 3: In the Combine section click Merge Queries.



Step 4: Choose Merge Queries to merge directly into the selected table or Merge Queries as New to create a new merged table.

Step 5: From the first drop-down select sales Data and choose the common column

Step 6: From the second drop-down select product and choose the matching column.

Merge

Select tables and matching columns to create a merged table.

sales2019

order_id	product_id	store_id	order_date	order_date_2	sales	revenue	stock	price	prom
40911	P0392	S0001	1/1/2019	1-Jan-19	0	0	3	124.99	PR14
40912	P0018	S0071	1/1/2019	1-Jan-19	0	0	4	3.25	PR03
40913	P0565	S0096	1/1/2019	1-Jan-19	0	0	12	4.25	PR14
40914	P0372	S0121	1/1/2019	1-Jan-19	0	0	24	4.9	PR14

product

product_id	product (brand)	type	length	depth	width	cate
P0000	Serum (Livon)	Indian & Exotic Herbs	1.968505	20	12	Fruits & V
P0001	Hand Wash - Moisture Shield (Savlon)	Hair Oil & Serum	5.3149635	22	20	Beauty &
P0002	Good Day Butter Cookies (Britannia)	Hand Wash & Sanitizers	8.661422	40	22	Beauty &
P0004	Happy Happy Choco-Chip Cookies (Parle)	Cookies	0.787402	13	4	Snacks &

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

› Fuzzy matching options

✓ The selection matches 47239 of 47239 rows from the first table.

OK Cancel

Step 7: Click OK. The matching data from sales will be added to Product based on the selected key.

A ^B _C order_id	A ^B _C product_id	A ^B _C store_id	A ^B _C order_date	A ^B _C order_date_2
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
40911	P0392	S0001	1/1/2019	1-Jan-19
40912	P0018	S0071	1/1/2019	1-Jan-19
40913	P0565	S0096	1/1/2019	1-Jan-19
40914	P0372	S0121	1/1/2019	1-Jan-19
40915	P0543	S0101	1/1/2019	1-Jan-19
40916	P0286	S0046	1/1/2019	1-Jan-19
40917	P0428	S0085	1/1/2019	1-Jan-19
40918	P0171	S0079	1/1/2019	1-Jan-19
40919	P0125	S0084	1/1/2019	1-Jan-19
40920	P0500	S0103	1/1/2019	1-Jan-19
40921	P0399	S0125	1/1/2019	1-Jan-19
40922	P0499	S0093	1/1/2019	1-Jan-19
40923	P0663	S0058	1/1/2019	1-Jan-19

☑ Example

Table 1: Students

StudentID Name

1	A
2	B

Table 2: Marks

StudentID Marks

1	80
2	85

After Merge →

StudentID Name Marks

1	A	80
2	B	85

Join Types

- ❖ **Left Outer Join:** Includes all rows from the first table and matching rows from the second table.

A **Left Outer Join** returns:

- ☑ All records from the **left (first) table**
- ✚ Matching records from the **right (second) table**
- ✗ If no match → NULL (blank) values from the right table

Step-by-Step: Left Outer Join

Step 1: Open Power Query

1. Open Power BI Desktop
2. Click **Home** → **Transform Data**

Step 2: Start Merge

1. Select the **Left Table**
2. Go to **Home** tab
3. Click **Merge Queries**

Step 3: Select Tables

1. Choose the **Right Table**

2. Select the matching column in both tables (e.g., StudentID)

Step 4: Choose Join Type

From dropdown select:

☒ **Left Outer (all from first, matching from second)**

Click **OK**

Step 5: Expand Columns

1. Click the expand icon ($\Rightarrow$)
2. Select required columns
3. Click **OK**

☒ **Example**

Left Table: Students

StudentID Name

1	A
2	B
3	C

Right Table: Marks

StudentID Marks

1	80
2	85

Result After Left Outer Join

StudentID Name Marks

1	A	80
2	B	85
3	C	NULL

- ❖ **Right Outer Join:** Includes all rows from the second table and matching rows from the first table.

A **Right Outer Join** returns:

- ☒ All records from the **right (second) table**
- +** Matching records from the **left (first) table**
- ×** If no match $\rightarrow$ NULL (blank) values from the left table

Step-by-Step: Right Outer Join

Step 1: Open Power Query

1. Open Power BI Desktop
2. Click **Home** → **Transform Data**

Step 2: Start Merge

1. Select the **Left Table**
2. Click **Home** → **Merge Queries**

Step 3: Select Matching Columns

1. Choose the **Right Table**
2. Select common column in both tables (e.g., StudentID)

Step 4: Choose Join Type

From dropdown select:

☒ **Right Outer (all from second, matching from first)**

Click **OK**

Step 5: Expand Columns

1. Click the expand icon
2. Select required columns
3. Click **OK**

☒ **Example**

Left Table: Students

StudentID Name

1	A
2	B

Right Table: Marks	StudentID	Marks
	1	80
	2	85
	3	90

Result After Right Outer Join	StudentID	Name	Marks
	1	A	80
	2	B	85
	3	NULL	90

❖ **Full Outer Join:** Includes all rows from both tables.

A **Full Outer Join** returns:

- ☒ All records from the **left table**
- ☒ All records from the **right table**
- +** Matching records are combined
- ×** Non-matching records show **NULL (blank)** values

◇ Step-by-Step: Full Outer Join

Step 1: Open Power Query

1. Open Power BI Desktop
2. Click **Home** → **Transform Data**

Step 2: Start Merge

1. Select first table
2. Click **Home** → **Merge Queries**

Step 3: Select Tables & Columns

1. Choose second table
2. Select common column in both tables (e.g., StudentID)

Step 4: Choose Join Type

From dropdown select:

☒ **Full Outer** (all rows from both)

Click **OK**

Step 5: Expand Columns

1. Click expand icon
2. Select required columns
3. Click **OK**

☒ Example

Left Table: Students

StudentID Name

1	A
2	B

Right Table: Marks

StudentID Marks

2	85
3	90

Result After Full Outer Join

StudentID Name Marks

1	A	NULL
2	B	85
3	NULL	90

❖ **Inner Join:** Includes only matching rows from both tables.

An **Inner Join** returns:

- ☒ Only the records that **match in both tables**
- ☒ Non-matching records from either table are removed

Step-by-Step: Inner Join

Step 1: Open Power Query

1. Open Power BI Desktop
2. Click **Home** → **Transform Data**

Step 2: Start Merge

1. Select the first table
2. Click **Home** → **Merge Queries**

Step 3: Select Tables & Columns

1. Choose second table
2. Select matching column in both tables (e.g., StudentID)

Step 4: Choose Join Type

From dropdown select:

☒ **Inner (only matching rows)**

Click **OK**

Step 5: Expand Columns

1. Click expand icon
2. Select required columns
3. Click **OK**

☒ **Example**

Left Table: Students

StudentID Name

1	A
2	B
3	C

Right Table: Marks

StudentID Marks

1	80
2	85
4	90

Result After Inner Join

StudentID Name Marks

1	A	80
2	B	85

- ❖ **Left Anti Join:** Includes only the rows from the first table that don't have matching rows in the second table.

A **Left Anti Join** returns:

- ☒ Only the rows from the **left (first) table**
- ☒ That **do NOT have a match** in the right (second) table

Step-by-Step: Left Anti Join

Step 1: Open Power Query

1. Open Power BI Desktop
2. Click **Home** → **Transform Data**

Step 2: Start Merge

1. Select the **Left Table**
2. Click **Home** → **Merge Queries**

Step 3: Select Tables & Columns

1. Choose the **Right Table**
2. Select matching column in both tables (e.g., StudentID)

Step 4: Choose Join Type

From dropdown select:

- ☒ **Left Anti (rows only in first)**

Click **OK**

- ☒ **Example**

Left Table: Students

StudentID Name

1	A
2	B
3	C

Right Table: Marks

StudentID Marks

1	80
2	85

	StudentID	Name
Result After Left Anti Join	3	C

- ❖ **Right Anti Join:** Includes only the rows from the second table that don't have matching rows in the first table.

A **Right Anti Join** returns:

- ☒ Only the rows from the **right (second) table**
- ☒ That **do NOT have a match** in the left (first) table

Step-by-Step: Right Anti Join

Step 1: Open Power Query

1. Open Power BI Desktop
2. Click **Home** → **Transform Data**

Step 2: Start Merge

1. Select the **Left Table**
2. Click **Home** → **Merge Queries**

Step 3: Select Tables & Columns

1. Choose the **Right Table**
2. Select matching column in both tables (e.g., StudentID)

Step 4: Choose Join Type

From dropdown select:

- ☒ **Right Anti (rows only in second)**

Click **OK**

◆ **Output**

Power BI will directly show only unmatched rows from the right table.

☒ Example

Left Table: Students

StudentID Name

1	A
2	B

Right Table: Marks

StudentID Marks

1	80
2	85
3	90

Result After Right Anti Join

StudentID Marks

3	90
---	----

 Complete Comparison Table

Join Type	Keeps All Left	Keeps All Right	Keeps Only Matching	Keeps Unmatched
Inner Join	✗	✗	☑	✗
Left Outer Join	☑	✗	☑	Left only
Right Outer Join	✗	☑	☑	Right only
Full Outer Join	☑	☑	☑	Both sides
Left Anti Join	✗	✗	✗	Left only
Right Anti Join	✗	✗	✗	Right only

Join Kind

Left Outer (all from first, matching from second) ▼

Left Outer (all from first, matching from second)
 Right Outer (all from second, matching from first)
 Full Outer (all rows from both)
 Inner (only matching rows)
 Left Anti (rows only in first)
 Right Anti (rows only in second)

Step-by-Step: Merge Queries

1. Open **Power BI Desktop**
2. Click **Home** → **Transform Data**
3. In Power Query Editor → Select first table
4. Click **Home tab** → **Merge Queries**
5. Select second table
6. Select matching column in both tables
7. Choose **Join Type**:
 - Left Outer (most common)
 - Right Outer
 - Inner
 - Full Outer
8. Click **OK**
9. Expand the merged column to select required fields

☒ Example:

Table 1: Students

StudentID	Name
1	A
2	B

Table 2: Marks

StudentID	Marks
1	80
2	85

After Merge →

StudentID	Name	Marks
1	A	80
2	B	85

Append Queries

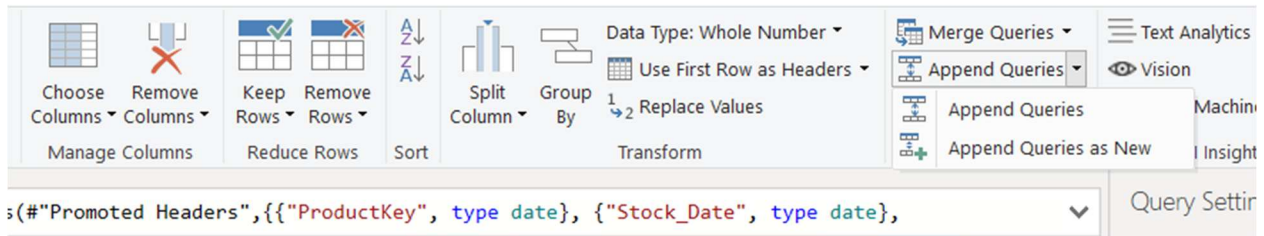
Append Queries are used to combine multiple tables by stacking them vertically, adding rows from one table to another. When the column structure matches data from the appended table is placed below the existing data in the base table.

- ❖ Works vertically by adding new rows
- ❖ Keeps the same number of columns in the base table
- ❖ Combines two or more queries into a single table
- ❖ Requires matching column names and structure

How to Append Queries

Step 1: In the Power Query Editor, select the query (table) that we want the other query to append to. For example, choose Sales Data.

Step 2: After selecting Sales Data Table click the Home Tab to access the Ribbon Menu.



Step 3: Select Append Queries from the Combine menu.

Step 4: Select Add Queries as New. A pop-up menu will appear.

Step 5: Pick first query from the first drop-down selection.

Step 6: Choose second query from the second drop-down selection

Append

Concatenate rows from two tables into a single table.

☒ Two tables ☐ Three or more tables

First table

sales2019

Second table

sales2018

OK

Cancel

Step 7: Once selected click OK to append the queries.

Queries [8]

- sales2017
- store
- product
- sales2019
- sales2018
- Merge1
- Merge2
- Append1

Table.Combine({sales2019, sales2018})

	order_id	product_id	store_id	order_date
1	40911	P0392	S0001	1/1/2019
2	40912	P0018	S0071	1/1/2019
3	40913	P0565	S0096	1/1/2019
4	40914	P0372	S0121	1/1/2019
5	40915	P0543	S0101	1/1/2019
6	40916	P0286	S0046	1/1/2019
7	40917	P0428	S0085	1/1/2019
8	40918	P0171	S0079	1/1/2019
9	40919	P0125	S0084	1/1/2019
10	40920	P0500	S0103	1/1/2019

Example

Table 1	Name Sales
---------	------------

A 100

Table 2

Name Sales

B 200

After Append →

Name	Sales
------	-------

A 100

B 200

Step-by-Step: Append Queries

1. Open Power Query Editor
2. Go to **Home** tab
3. Click **Append Queries**
4. Select:
 - o Two tables OR
 - o Three or more tables
5. Click **OK**

Difference between Merge and Append Query

Feature	Merge Query	Append Query
Purpose	Combines two tables based on a shared column.	Stacks two or more tables vertically, combining them into one.
Requirement	At least one matching column must exist between the tables.	Tables to be appended should have the same columns.
Number of Columns	The number of columns may differ between the queries.	All tables must have the same number of columns to be appended.
Result	Adds new columns to the existing query.	Adds more rows to the existing query.
When to Use	When combining data from two tables based on a common column or creating new relationships.	When adding more rows of data to an existing table
Purpose	Combines two tables based on a shared column.	Stacks two or more tables vertically, combining them into one.

S.No	Questions
01	Define Power Query? List and explain different Data types and Transformations of Power Query.
02	Explain different basic steps required Power Query
03	List and explain different types PBI filters.
04	Explain how to add Power BI Filter to a Visual?
05	Explain how to add Power BI Filter to a Page?
06	Explain how to add Power BI Filter to a report?
07	List and explain steps involve in power pivot operation with example.
08	List and explain steps involve in power unpivot operation with example.
09	List and explain steps involve in power transpose operation with example.
10	List and explain steps involve in merge Queries operation with example.
11	List and explain different types of Power BI join
12	List difference between different types of joins
13	List and explain steps involve in append Queries operation with example.
14	List difference between Merge and Append Query